Auditable Unlearning for Retrieval-Augmented and Fine-Tuned Foundation Models

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Abstract—Deletion requests are easy to honor on paper and hard to verify in practice. In a modern foundation-model application a deleted record rarely lives in the model weights alone: it can persist through retrievers, vector indexes, summaries, adapters, caches, and generated traces, so a system can pass a weight-level unlearning test while still behaving as if nothing was removed. We argue that deletion should instead be verified as *counterfactual behavioral indistinguishability*—a post-deletion system should behave like a clean reference system, S_{clean}, rebuilt as if the record had never entered the pipeline—and we make this property testable. ADCU lifts (ε, δ) deletion from model-centric removal to compositional RAG and fine-tuned systems: it scores residual risk as a two-sample distance to S_{clean}, allocates a black-box probe budget over provenance-linked target–channel arms under a declared probe-coverage assumption, and uses six diagnostic channels to localize where residual dependence is mediated. The companion benchmark, ADCU-Bench, spans RAG deletion, fine-tuning-record unlearning, hybrid RAG-to-fine-tuning retention, dense retrieval, TOFU-style forget/retain evaluation, clean-counterfactual testing, and causal component interventions. Across 6,840 repeated audit rows, ADCU flags every configured RealRAG, NaturalFEVER, and HybridReal failure with zero false alarms at the declared operating point, and a 5,000-record scaling check shows the effect is not an artifact of tiny corpora; in contrast, exact-match, canary-only, retriever-only, membership-style, and extraction-only audits each miss entire residual-dependence channels.

Impact Statement— Foundation-model systems are increasingly used in settings where users, organizations, and regulators may require deletion of contributed data. Yet deletion claims are hard to trust when a record can survive through retrieval indexes, summaries, caches, adapters, or derivatives. This work advances trustworthy AI by turning deletion verification into a measurable behavioral audit for RAG and fine-tuned foundation models. The proposed metric, algorithm, benchmark, and attack suite help practitioners identify residual dependence, compare unlearning procedures, preserve retain utility, and document uncertainty through confidence-bounded decisions. The broader impact is a practical assurance layer for AI governance: rather than relying on informal claims that a model or pipeline has forgotten, operators can produce repeatable, redacted, and technically interpretable evidence about what was tested and where residual deletion risk remains.

Index Terms—Trustworthy artificial intelligence, machine unlearning, retrieval-augmented generation, foundation models, data provenance, behavioral auditing.

1 Introduction

A modern foundation-model application is no longer a single model queried in isolation: a deployed assistant or enterprise analyst tool stitches retrieval-augmented generation (RAG), dense and lexical indexes, rerankers, supervised fine-tuning and LoRA adapters, -data generation, and caches into one pipeline. RAG connects language models to external knowledge sources [1], [2], [3], while instruction tuning and adapter-based specialization fold private or domain-specific records directly into model behavior. This improves utility but creates a core trustworthy-AI problem: after a deletion request, an operator can remove the original record while the deployed system keeps behaving as if it remained—through retrieved chunks, embeddings, summaries, question–answer pairs, cache entries, adapters, or evaluation traces.

Privacy and governance regimes motivate data erasure and deletion accountability [5], [7], and machine unlearning provides the algorithmic foundation for making systems forget selected data [8], [9], [10], [11], [13], [14], [16]. But in modern LLM and RAG deployments the deletion problem is not confined to the parameter vector: a system can pass a model-centric unlearning test while still retrieving a stale chunk, citing a shadow copy, answering from a derivative, or paraphrasing a fact learned through an adapter. Extraction and membership-inference work shows that such dependence can be exposed through black-box interaction [17], [18], [19], [20], [21], [22], and RAG evaluation studies retrieval quality and attribution [24]; yet none directly answers the operator's audit question: after a deletion request, what measurable evidence shows that the deleted data and its downstream derivatives no longer influence retrieval, generation, or fine-tuned behavior?

We argue that deletion verification should be treated as *counterfactual behavioral indistinguishability*. The relevant question is not whether a chosen leakage string appears, but whether the post-deletion system behaves like a clean reference system S_{clean} rebuilt as if the deleted record had never entered any retriever, cache, -data generator, adapter, or evaluation trace. This avoids circularity: diagnostic channels can localize failures, but the gold standard is statistical distance from the clean counterfactual. A practical audit then needs three ingredients: provenance (which artifacts are downstream of the target), efficient adaptive testing over high-risk target–channel arms, and behavioral tests that compare S_- with S_{clean} and explain whether the residual distance is retrieval-, parametric-, cache-, or derivative-mediated.

This article proposes ADCU (*Auditable Data-Centric Unlearning*), a framework for deletion-risk auditing in RAG and fine-tuned foundation-model systems. ADCU represents deletion targets and their downstream artifacts in a provenance graph; defines an (ε, δ) behavioral-deletion guarantee as bounded distance between S_- and S_{clean} over an adversary-aware probe class; allocates probes through a graph-structured sequential test; and uses direct, paraphrase, retrieval, counterfactual, extraction, trigger, and graph-pivot probes as diagnostic localizers rather than as the failure definition. Instead of a binary claim of perfect forgetting, it returns a confidence-bounded pass, fail, or escalate decision under an operator-chosen tolerance. The goal is not to replace unlearning algorithms, certified removal, differential privacy, or legal review [27], [11], [40], but to provide measurable technical evidence about residual behavioral dependence in the pipeline surrounding those mechanisms.

We instantiate the framework in ADCU-Bench, spanning four core evaluation tracks:

- *RealRAG* [?]: Tests RAG deletion by loading the FEVER dataset as a public retain corpus, injecting private targets, and evaluating whether provenance-guided purges succeed while index-only or cache-forgetful deletions fail.
- *NaturalFEVER* [?]: Audits natural data deletion by treating public FEVER passages themselves as deletion units and testing whether raw evidence, citation caches, or answer-only paraphrases still influence responses after removal.
- *FineTuneSim*: A controlled adapter-memory track that models residual parameter-side memory, testing unlearning methods like gradient-ascent and negative fine-tuning against over-unlearning and replacement failures.
- *HybridReal*: A hybrid track that stresses both retrieval and fine-tuning pipelines by combining RAG retention with derivatives that survive through summaries, cache artifacts, graph-pivot expansion, or model-side instruction examples.

We further add LoRA-SFT [?] and cached PEFT/LoRA validations, pretrained LoRA margin analysis, TOFU-style forget/retain evaluation , neural dense retrieval (MiniLM/E5/BGE), clean-counterfactual two-sample tests, and component interventions for causal mediation. Across the repeated harness, ADCU detects configured deletion failures in RealRAG, NaturalFEVER, and HybridReal with zero configured false alarms; in FineTuneSim it detects 72.2% of all configured failures, or 96.3% of residual adapter-memory failures once utility-only over-unlearning cases are separated as escalations. The results do not prove legal compliance or absolute absence of influence; they show that deletion audits become substantially more informative when they are clean-counterfactual, provenance-aware, graph-adaptive, and multi-channel.

This article makes the following contributions:

- We formulate deletion verification for RAG and fine-tuned foundation-model systems as (ε, δ) counterfactual behavioral indistinguishability from a clean reference pipeline.
- We define deletion risk as clean-counterfactual behavioral distance and use direct, paraphrase, retrieval, extraction, and watermark evidence as diagnostic localizers rather than circular ground truth.
- We present a graph-structured adaptive audit algorithm with confidence bounds over provenance-linked target–channel arms, plus causal intervention diagnostics that decompose residual risk into retrieval-, cache-, derivative-, and parametric-mediated components.
- We introduce ADCU-Bench, a reusable benchmark with four core evaluation tracks (*RealRAG* [?], *NaturalFEVER* [?], *FineTuneSim*, *HybridReal*) spanning private deletion, natural public-evidence deletion, RAG retention, fine-tuning residues, hybrid derivative retention, dense retrieval, LoRA-SFT [?] and PEFT/LoRA validations, pretrained LoRA margin analysis, and TOFU-style forget/retain evaluation , with an attack and baseline suite showing where exact-match, canary-only, retriever-only, membership-inference, extraction-only, and influence-proxy audits fail.

Section 2 reviews related work; Section 3 defines the audit problem and threat model; Sections 4–5 present the deletion-risk metric and audit algorithm; Section 6 describes ADCU-Bench; Section 7 reports experiments; Sections 8–9 discuss limitations and reproducibility; Section 10 concludes.

2 Related Work

Trustworthy auditing and data governance. Trustworthy AI evaluation increasingly requires evidence about deployed behavior rather than only benchmark accuracy, and foundation-model deletion is a hard audit target because residual dependence can appear through retrieval, adaptation, memory, or generated derivatives. Recent challenge work on federated foundation models lists private-data utilization, continual learning, unlearning, and watermarking among the major open problems [7]. ADCU contributes to this agenda by making deletion claims testable through declared probes, confidence bounds, and artifact-level provenance; unlike explainability or attribution work, it asks whether a specific deleted source still contributes to post-deletion behavior, not merely which context was salient for one answer.

Machine unlearning. Machine unlearning removes the effect of selected training data after deployment [8], [10], [9], and foundation-model unlearning extends this to large pretrained and adapted models where exact retraining is infeasible [13], [14]. Benchmarks such as TOFU, MUSE, RWKU, and WMDP evaluate fictitious-biography unlearning, six-way unlearning properties, real-world knowledge removal, and hazardous-knowledge reduction , and surveys emphasize evaluating both forgetting and retained utility [16]. We differ by focusing on auditability across the data pipeline rather than proposing a new weight-update rule: ADCU can wrap a TOFU-, MUSE-, RWKU-, or WMDP-style run and ask whether downstream artifacts, caches, derivatives, or adapters remain behaviorally distinguishable from a clean reference pipeline.

Retrieval-augmented generation and provenance. RAG conditions generation on retrieved documents [1], [2], [3], which improves grounding but introduces deletion risk: removing a source from one index need not remove derived chunks, summaries, cached contexts, or fine-tuning examples, and multi-stage pipelines can have a clean first-stage index while a reranker, cache, or derivative reintroduces a deleted fact. Work on attribution, context usage, source tracing, and RAG evaluation [24] treats retrieval quality as the object of study; ADCU instead treats these artifacts as first-class audit objects and adds deletion risk as an orthogonal axis, motivating the DenseRAG extension over lexical, dense SVD, neural, and cross-encoder reranking under one audit interface.

Data valuation and influence. Data-valuation methods estimate how much individual records contribute to utility [32], [33], [34], [35], and influence functions estimate the effect of training examples on predictions [36], [37], [38], [39]. ADCU uses this line differently: valuation prioritizes audit effort after a deletion request rather than scoring training contributions.

Extraction, membership, canaries, and watermarking. Language models can expose training data through extraction [18], [17], [19], membership inference [20], [21], [22], and canary or watermark signals [40]. These black-box channels reveal data dependence even when accuracy looks acceptable, but each is incomplete: canary-only checks fail once the marker is suppressed, membership tests miss retrieval-only leakage, and extraction prompts miss ordinary paraphrased dependence. Our benchmark therefore includes membership-inference, extraction, retriever-only, and influence-proxy baselines beside full ADCU, and incorporates triggered-leakage, graph-pivot, retrieval-dependence, and watermark evidence rather than relying on any single axis.

Privacy, compliance, and memory governance. Deletion auditing is motivated by data-erasure rights, but technical audits should not overclaim legal compliance; prior work on differential privacy, certified removal, and deletion-as-control provides context [27], [5], [11]. ADCU contributes measurable evidence that coexists with such mechanisms: when certified removal or DP continual release is available, it monitors pipeline drift and derivative artifacts outside the formal core. Finally, adaptive-retrieval gating motivates two diagnostics in ADCU-Bench: an answer-only black-box audit for deployments that hide retrieval metadata, and a retrieval-off counterfactual audit that tests whether disabling retrieval changes the residual-dependence score.

3 Problem Formulation

Let S_0 be the pre-deletion system and S_- be the post-deletion system. A system may include a base model, retriever, vector index, reranker, prompt template, adapter, cache, and operational logs. A deletion request targets a set of data units $D_{\text{del}} = \{z_1, \ldots, z_k\}$. Each target may correspond to raw records, document chunks, embeddings, derivatives, or fine-tuning examples.

The counterfactual gold standard is a clean reference system $S_{\text{clean}}(D_{\text{del}})$ built as if the deletion targets had never entered any pipeline component: no raw record, chunk, embedding, cache item, derivative, adapter update, evaluation trace, or summary is produced from D_{del}. Exact construction of S_{clean} may require retraining, rebuilding indexes, regenerating data, and purging caches. When that construction is infeasible, it remains the formal oracle that defines deletion success; practical audits estimate distance to this oracle.

We model data dependence with a provenance graph $G = (V, E)$. Nodes include deletion targets and downstream artifacts. An edge (u, v) means artifact v was produced from or influenced by u. Examples include raw-document-to-chunk, chunk-to-embedding, document-to-summary, summary-to–example, and -example-to-adapter edges.

Behavioral-deletion guarantee. Let $\mathcal{Q}(D_{\text{del}}, G)$ be a probe class and let $O_S(q)$ denote the observable behavior of system S on probe q, including generated answer, retrieved context identifiers, citations, and available metadata. For a bounded discrepancy $d(\cdot,\cdot) \in [0,1]$, define

$$\Delta_{\mathcal{Q}}(S_-, S_{\text{clean}}) = \sup_{q \in \mathcal{Q}} d\big(P(O_{S_-}(q)), P(O_{S_{\text{clean}}}(q))\big)\,.$$

We say S_- satisfies (ε, δ) behavioral deletion for D_{del} over $\mathcal{Q}$ if, with probability at least $1-\delta$ over audit sampling and system randomness,

$$\Delta_{\mathcal{Q}}(S_-, S_{\text{clean}}) \le \varepsilon.$$

This definition lifts counterfactual indistinguishability from model-centric unlearning to compositional AI pipelines: a post-deletion system must behave like the system that would have existed had the record never affected retrieval, caches, derivatives, or adapters.

The audit problem is to test whether S_- is behaviorally indistinguishable from S_{clean} while preserving utility on a retain distribution Q_{ret}. We do not require the auditor to inspect model weights. Instead, the auditor can submit black-box probes and observe generated answers, retrieved contexts, citations, and available metadata. The six evidence channels used later are estimators and diagnostics for this counterfactual distance; they are not the definition of deletion failure.

An audit returns one of three decisions. *Pass* means the floor-normalized excess counterfactual risk is below the declared operating tolerance and retain utility is acceptable. *Fail* means the post-deletion system is distinguishable from the clean counterfactual or a diagnostic channel localizes a violation. *Escalate* means the evidence is inconclusive, the clean reference is unavailable, or retain utility has collapsed.

Threat model. The auditor may face incomplete deletion operations: raw chunks may be removed while shadow copies, summaries, examples, caches, or adapters remain active. The baseline attacker class is black-box or gray-box: the auditor queries the post-deletion application and, in gray-box RAG settings, inspects retrieved context identifiers and citations, but does not assume access to model weights or training gradients. A white-box self-influence adversary with access to gradients, adapter weights, or internal memorization scores is outside the guarantee boundary by assumption; such tools can be added as privileged evidence. We do not claim that black-box auditing proves absolute absence of influence under all prompts; the objective is

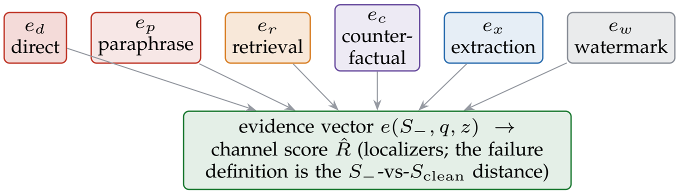

Fig. 1. The six diagnostic evidence channels. Each fires under a different failure mode—direct/paraphrase under model-side leakage, retrieval under stale or shadow-copy artifacts, counterfactual under any S_--vs-S_{clean} divergence, extraction under adversarial probes, watermark under canary or provenance-token hits—and together they localize residual dependence without themselves defining deletion failure.

bounded evidence of counterfactual behavioral indistinguishability over a declared, adversary-aware probe class.

4 Behavioral Deletion Risk

ADCU defines deletion risk as distance from the clean counterfactual, not as the mere presence of a hand-written leakage pattern. For a target z, let $A(z, G)$ be a target- and graph-conditioned audit distribution over probes in $\mathcal{Q}$. Let $Y_-(q) = O_{S_-}(q)$ and $Y_c(q) = O_{S_{\mathrm{clean}}}(q)$. Given a bounded behavioral discrepancy $d(Y_-, Y_c) \in [0, 1]$, the target-level risk is

$$R_{\mathrm{cf}}(z; S_-, S_{\mathrm{clean}}) = \mathbb{E}_{q \sim A(z,G)} \left[d(Y_-(q), Y_c(q)) \right].$$

The request-level risk is

$$R_{ADCU}(D_{\mathrm{del}}) = \sum_{z \in D_{\mathrm{del}}} \pi(z) R_{\mathrm{cf}}(z; S_-, S_{\mathrm{clean}}),$$

where $\pi(z)$ is an audit prior induced by target influence and provenance degree. In deterministic deployments, d can be a normalized answer-and-retrieval distance. In stochastic deployments, d can be a two-sample distance between output distributions, such as total variation over discretized answer features, maximum mean discrepancy over embeddings, or a calibrated classifier's distinguishability advantage. The audit reports the raw UCB and applies the floor-normalized operating rule below so finite-budget clean systems are not rejected solely by the confidence floor; retain utility must also remain above threshold.

Diagnostic evidence channels. The released benchmark also computes an evidence vector

$$e(S_-, q, z) = [e_d, e_p, e_r, e_c, e_x, e_w],$$

where e_d measures direct leakage, e_p paraphrased leakage, e_r retrieval dependence, e_c clean-counterfactual answer dependence, e_x extraction risk, and e_w watermark or provenance-token hits. These channels are not the ground-truth definition of deletion failure. They are transparent estimators and localizers for the counterfactual distance: when S_- differs from S_{clean}, the channel vector (Fig. 1) indicates whether the difference is direct, semantic, retrieval-mediated, extraction-mediated, or provenance-token-mediated.

Operational scorers. The artifact uses deterministic, versioned scorers so audit rows are reproducible. Direct leakage is normalized token overlap with protected text. Paraphrase leakage is maximum overlap with aliases, protected facts, and probe-specific expected terms. Retrieval dependence is one when the response cites or retrieves an artifact reachable from the deleted target in G and the clean reference does not. Counterfactual answer dependence compares the answer under S_- with the clean response and fires when the deleted target changes the answer semantics or retrieval evidence. Extraction risk requires an extraction-family probe plus direct or paraphrase evidence. Watermark evidence is an exact canary or provenance-token hit. Deployments can replace these scorers with frozen embedding, NLI, or judge models, but must report judge versions, thresholds, and calibration data.

Two-sample audit statistic. For the minimal black-box instantiation, sample n_z probes $q_i \sim A(z, G)$, query both S_- and S_{clean}, and compute

$$\hat{R}_{\mathrm{cf}}(z) = \frac{1}{n_z} \sum_{i=1}^{n_z} d(O_{S_-}(q_i), O_{S_{\mathrm{clean}}}(q_i)).$$

Because $d \in [0, 1]$, ADCU reports

$$\mathrm{UCB}_{\mathrm{cf}}(z) = \hat{R}_{\mathrm{cf}}(z) + \sqrt{\frac{\log(1/\delta_z)}{2n_z}}.$$

The square-root term creates an irreducible confidence floor even when the empirical clean-counterfactual distance is zero. We therefore report the raw UCB and use a floor-corrected excess risk for finite-budget operating decisions:

$$\widetilde{R}_{\mathrm{cf}}(z) = \frac{\max\{0, \mathrm{UCB}_{\mathrm{cf}}(z) - r_0(n_z, \delta_z)\}}{1 - r_0(n_z, \delta_z)},$$
$$r_0(n_z, \delta_z) = \sqrt{\frac{\log(1/\delta_z)}{2n_z}}.$$

The excess score is not a new guarantee; it is a calibrated way to state how far the observed risk rises above the clean-UCB floor. A floor-normalized operating tolerance τ_{ex} corresponds to the raw-UCB condition $\mathrm{UCB}_{\mathrm{cf}}(z) \le r_0 + (1 - r_0)\tau_{\mathrm{ex}}$. The experiments report both raw $\mathrm{UCB}_{\mathrm{cf}}$ and $\widetilde{R}_{\mathrm{cf}}$, and the headline pass/fail rule uses the declared excess-risk tolerance so clean systems with zero empirical counterfactual distance are not rejected only because the audit budget is finite. If S_{clean} is expensive, the audit can use an archived clean index, filtered retraining on a small slice, or a clean simulator for benchmark evaluation. The article reports both the counterfactual statistic when the clean reference is available and the channel scores used to diagnose why a system is distinguishable.

Proposition 1 (Counterfactual false-pass control)**.** *Assume $d(O_{S_-}(q), O_{S_{\mathrm{clean}}}(q)) \in [0, 1]$ and probes are sampled independently from $A(z, G)$. If the audit passes target z only when $\mathrm{UCB}_{\mathrm{cf}}(z) \le \varepsilon$, then*

$$\Pr[\mathrm{pass}(z) \wedge R_{\mathrm{cf}}(z) > \varepsilon] \le \delta_z.$$

Proof sketch. Hoeffding's inequality bounds the probability that the true mean counterfactual distance exceeds $\hat{R}_{\mathrm{cf}}(z) + \sqrt{\log(1/\delta_z)/(2n_z)}$ by δ_z. The pass event requires this upper bound to be no larger than ε, so passing while the true risk exceeds ε is contained in the Hoeffding failure event.

Proposition 2 (Certified graph-structured adaptive audit)**.** *Let each arm be a target–channel pair $a = (z, c)$ and let $N(a)$ be the set of provenance-neighbor arms sharing a downstream artifact, cache, adapter, or derivative family. For every neighbor edge $b \to a$, assume a scorer contract provides a bounded pseudo-observation $Y_i^{a \leftarrow b} \in [0, 1]$ with $\mathbb{E}[Y_i^{a \leftarrow b} \mid b \text{ sampled}] = r(a)$, and assign it weight $\rho_{ab} \in [0, 1]$. Let $\hat{r}_t(a)$ be the weighted mean of direct observations from a and certified pseudo-observations from neighbors, with effective sample size*

$$n_t^{\text{eff}}(a) = n_t(a) + \sum_{b \in N(a)} \rho_{ab} n_t(b).$$

If the audit ranks arms by

$$B_t(a) = \hat{r}_t(a) + \sqrt{\frac{\log(2|\mathcal{A}|t^2/\delta)}{2n_t^{\text{eff}}(a)}},$$

then, with probability at least $1 - \delta$, all certified arm risks are simultaneously upper-bounded by $B_t(a)$ for all audit times t.

Proof sketch. Under the scorer contract, direct observations and certified neighbor pseudo-observations are bounded estimates of the same arm mean $r(a)$; the weights define a standard weighted bounded-mean estimator. Weighted Hoeffding gives the displayed radius up to the conservative effective sample count above. Applying the bound to every arm and time with failure probability $\delta/(|\mathcal{A}|t^2)$ and union-bounding over arms and audit times gives simultaneous coverage. If this contract is unavailable, ADCU uses graph scores only to prioritize which arm to sample next; the formal false-pass certificate then falls back to direct clean-counterfactual samples and Proposition 1.

Proposition 3 (Probe coverage against bounded adversaries)**.** *Let $\mathcal{H}_{k,\eta}$ be an adversary class whose leakage can be elicited by either a direct/paraphrase probe within semantic radius η of a protected fact or by a k-hop provenance pivot from a deleted artifact. If the probe generator samples m independent paraphrase neighborhoods and graph pivots such that each adversarial leakage mode is hit with probability at least $p_{\min}$, then the probability that an active leakage mode in $\mathcal{H}_{k,\eta}$ is missed is at most $(1 - p_{\min})^m$. Conversely, any black-box audit with budget m and hit probability at most $p_{\min}$ for a hidden mode has miss probability at least $(1 - p_{\min})^m$ for that mode.*

This coverage statement is deliberately conditional on an explicit adversary class. It replaces the vague claim that probes are representative with a testable requirement: report the semantic neighborhood, graph-hop radius, hit probability estimate, and budget. Learned red-team probes can then be evaluated by whether they increase $p_{\min}$ or discover additional modes outside $\mathcal{H}_{k,\eta}$.

Violation rule versus UCB rule. The implementation records any high-scoring diagnostic probe as a localized violation so operators can immediately inspect the responsible artifact or component. The formal false-pass guarantee, however, is attached only to the UCB rule above. A single-probe fail is therefore a conservative escalation/fail heuristic for triage, not a separate statistical certificate. When the two disagree, ADCU reports the disagreement and treats the decision as fail or escalate rather than pass.

Algorithm 1 ADCU graph-adaptive counterfactual audit

Build provenance graph G and deletion target set D_{del}
Build or approximate clean counterfactual S_{clean}
Form target–channel arms $\mathcal{A} = \{(z, c)\}$ from G
Generate direct, paraphrase, retrieval, extraction, trigger, and graph-pivot probes
Initialize each target with a minimum clean-vs-post-deletion probe budget
while audit budget remains and some arm has UCB above tolerance **do**
  Select arm a_t with largest graph-adjusted upper-confidence score
  Query S_- and S_{clean} on the selected probe
  Update counterfactual distance and diagnostic evidence channels
  Propagate conservative effective samples to provenance-neighbor arms
end while
Estimate retain utility on Q_{ret}
Run component interventions: retrieval off, cache purged, adapter swapped, derivative store purged
if counterfactual UCB $> \varepsilon$ or a localized violation is observed **then**
  return fail with causal component attribution
else if counterfactual UCB $\leq \varepsilon$ and retain utility is acceptable **then**
  return pass
else
  return escalate
end if

5 ADCU Audit Algorithm

The audit is intentionally black-box or gray-box: it can be applied when the operator has access to system responses and retrieval metadata but not necessarily to training gradients or model weights. The key distinction from the earlier channel-only view is that ADCU compares S_- against S_{clean}. Channel scores explain *why* the systems differ; counterfactual distance defines *whether* deletion succeeds.

Graph-structured pure exploration. Each audit arm is a target–channel pair, such as $(z, \text{retrieval})$, $(z, \text{paraphrase})$, or $(z, \text{adapter})$. The provenance graph induces correlations among arms: a probe that retrieves a deleted summary may inform cache, -QA, and adapter arms derived from that summary. ADCU uses this graph in two modes. In the certified mode of Proposition 2, neighbor observations are shared only when the scorer contract supplies bounded pseudo-observations for the same arm mean. Otherwise, the graph is an allocation heuristic: arms with many high-risk derivatives receive more probes, but the formal pass certificate is computed from direct clean-counterfactual samples. This reframes valuation-guided auditing as graph-structured sequential testing rather than a softmax heuristic while avoiding an unjustified UCB claim.

Informal composition theorem. Let S^- be the post-deletion pipeline, S_{clean} the clean reference, and $S^\diamond$ the provenance-closed version of S^- obtained by additionally removing every derivative of the deleted record. If the parametric component is within ε_θ of its clean counterfactual,

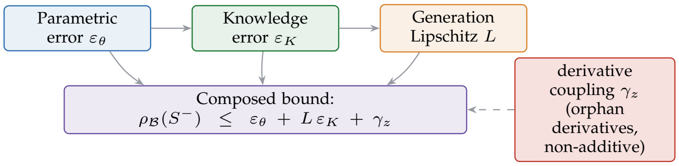

Fig. 2. Pipeline-deletion composition (Appendix A). Per-component deletion errors compose additively—ε_θ on parameters, ε_K on knowledge amplified by the generation Lipschitz factor L—plus a non-additive derivative-coupling term γ_z that is zero *iff* every derivative of the deleted record has also been removed; otherwise orphan derivatives dominate.

the retrieval/knowledge component is within ε_K, and generation is L-Lipschitz with respect to retrieval discrepancy, then the behavioral residual influence over probe class $\mathcal{B}$ obeys

$$\rho_{\mathcal{B}}(S^-) \le \varepsilon_\theta + L\varepsilon_K + \gamma_z, \quad \gamma_z = \sup_{q\in\mathcal{B}} \mathrm{TV}(P_{S^-}, P_{S^\diamond}).$$

The derivative-coupling term γ_z (Fig. 2) is zero exactly when deletion is provenance-closed: every chunk, embedding, summary, cache entry, example, and adapter-training derivative of the record has been removed or rendered behaviorally inert. Thus component-level unlearning guarantees compose only after lineage closure; otherwise orphan derivatives can dominate residual behavior. Appendix A gives the full theorem, multi-stage extension, and necessity proof.

Causal mediation of residual risk. After the distinguishability test finds residual risk, ADCU performs interventions on pipeline components. Let $S^{\mathrm{do}(R=0)}$ disable retrieval, $S^{\mathrm{do}(C=0)}$ purge caches, $S^{\mathrm{do}(A=0)}$ swap or remove adapters, and $S^{\mathrm{do}(D=0)}$ purge derivative stores. For component M, define the mediated residual contribution

$$\mathrm{Med}(M) = R_{\mathrm{cf}}(S_-, S_{\mathrm{clean}}) - R_{\mathrm{cf}}(S^{\mathrm{do}(M=0)}, S_{\mathrm{clean}}).$$

Under standard intervention assumptions–component toggles do not introduce new leakage, retain behavior outside the intervened path is stable, and probes are held fixed–$\mathrm{Med}(M)$ identifies how much distinguishability is mediated by that component. This turns retrieval-off ablations into a causal diagnostic: the audit can say whether deleted information survives primarily through retrieval artifacts, parametric adapter memory, caches, or derivatives.

Adaptive adversarial probes. Templated probes are reproducible but incomplete. ADCU therefore treats learned red-team probes as an optional budgeted arm: a generator proposes paraphrases within a declared semantic radius or k-hop graph pivots that maximize current evidence. The audit records whether learned probes discover violations missed by direct, paraphrase, retrieval, and extraction templates. This makes probe coverage an empirical object tied to the adversary class in Proposition 3 rather than an informal assumption.

6 ADCU-Bench

ADCU-Bench organizes deletion audits around three failure surfaces. A *RAG* track deletes source documents together with their chunks, embeddings, caches, and summaries; a *fine-tuning* track deletes instruction or preference records

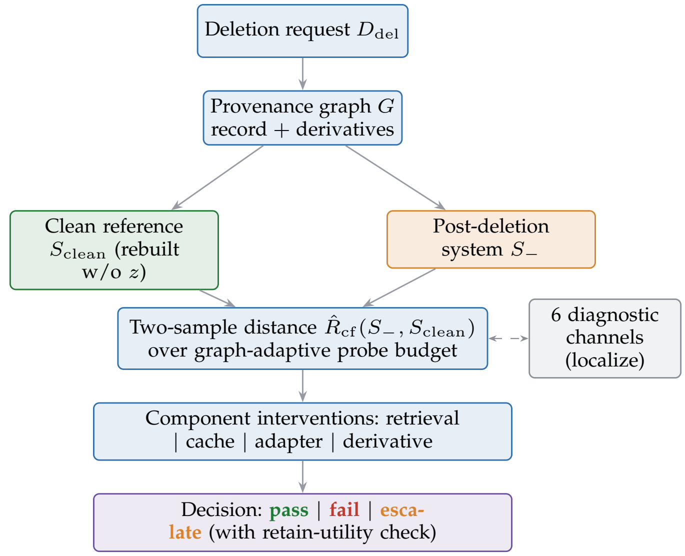

Fig. 3. ADCU audit pipeline. A deletion request is expanded through the provenance graph; the post-deletion system S_- is compared against the clean reference S_{clean} via a two-sample distance over a graph-adaptive probe budget, six diagnostic channels localize where any residual dependence is mediated, component interventions attribute it causally, and the audit returns a confidence-bounded decision with a retain-utility check.

that may have shaped an adapter; and a *hybrid* track stresses both at once—it generates instruction examples from RAG documents, fine-tunes an adapter on them, and then asks whether removing the original source actually removes the downstream behavior. Every track is instantiated in three states: the pre-deletion system S_0, the claimed post-deletion system S_-, and the clean reference S_{clean} rebuilt or simulated as if the target had never entered the pipeline.

Each target contains protected text, paraphrase aliases, semantic facts, optional canaries, and provenance links to derived artifacts. Each audit method is evaluated on counterfactual distinguishability from S_{clean}, failure detection, false alarms, risk score, residual-dependence channels, retain utility, and audit calls.

The benchmark includes weak audit baselines that mirror common but incomplete practices–exact-match, canary-only, and retriever-hit checks, uniform probes, and an ADCU ablation without graph-adaptive allocation–alongside membership-inference (direct and paraphrased answer dependence), extraction-audit (adversarial extraction probes), influence-proxy (a lightweight aggregate of the direct, paraphrase, and retrieval channels), and a two-sample clean-counterfactual baseline that queries S_- and S_{clean} on the same probes.

For causal localization, ADCU-Bench includes intervention variants that disable retrieval, purge caches, replace adapters, or remove derivative stores while holding probes fixed. The reported mediation scores decompose residual risk into retrieval-mediated, cache-mediated, derivative-mediated, and parametric components when the intervention assumptions in Section 5 hold.

To make the benchmark reusable, each run exports row-level JSON and CSV with seed, target size, audit budget, audit method, deletion method, decision, counterfactual distance, risk score, channel scores, intervention scores, retain utility, and call count. The benchmark card documents tracks, metrics, clean-reference construction, adversary class, probe coverage assumptions, and release assumptions. All private records used in the current artifact are , and case-study outputs are redacted.

7 EXPERIMENTS

We evaluate ADCU in two stages. First, we use a compact pilot to validate the metric, audit algorithm, benchmark schema, and attack suite. Second, we run a repeated submission-scale harness with a real-corpus FEVER RAG retain set, natural FEVER evidence deletion cases, simulated fine-tuning memories, and a hybrid RAG-to-derivative setting. The repeated harness spans five random seeds, three deletion-target sizes, three audit budgets, and the reviewer-response black-box/counterfactual modes, producing 6,840 audit-result rows.

Tracks. The *RealRAG* track loads FEVER [?] claims and evidence passages as the retain corpus, injects private deletion targets, and evaluates provenance-guided deletion, index-only deletion, shadow-copy deletion, cache-not-purged deletion, and backdoor-triggered RAG leakage. The *NaturalFEVER* track treats public FEVER [?] evidence passages themselves as deletion-request units, then audits whether raw evidence, citation caches, summaries, or answer-only paraphrases still influence behavior after removal. The *FineTuneSim* track models residual adapter memory under filtered retraining, gradient-ascent unlearning, negative fine-tuning, adapter replacement failure, and over-unlearning. The *HybridReal* track combines FEVER-style RAG retention with private derivatives that survive through examples, cache artifacts, graph-pivot expansion, or model-side behavior.

Table 1 summarizes repeated failure detection. The ADCU rows use the two-sample clean-counterfactual statistic $\hat{R}_{\mathrm{cf}}$ for pass/fail decisions; the other columns are diagnostic localizers. Full ADCU detects all configured RealRAG, NaturalFEVER, and HybridReal failures and 72.2% of FineTuneSim failures with zero false alarms under the conservative counterfactual operating point. Exact-match, canary-only, and retriever-hit audits each miss important channels: exact matching misses retrieval-only and paraphrase failures, canary-only auditing misses natural data without canaries and canary-suppression failures, and retriever-hit auditing misses model-side fine-tuning residues.

Table 2 reports the counterfactual statistic that underlies the revised definition. We use $\delta = 0.05$ and a floor-normalized excess-risk tolerance of 0.05; the raw UCB is reported beside the confidence floor so the operating rule is numerically auditable. Clean provenance-guided deletion has zero empirical clean-vs-post distance before the Hoeffding floor, while configured failures have positive $\hat{R}_{\mathrm{cf}}$ values. Under matched budgets (supplement), graph-adaptive allocation most improves FineTuneSim, where residual adapter memory is concentrated in fewer target–channel arms, and matches uniform allocation on the easier RAG and hybrid failures.

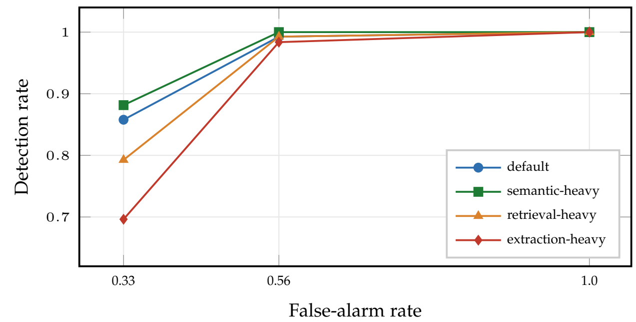

Fig. 4. Detection vs. false alarm across weight profiles as the operating tolerance is swept (Table 14). Lowering the tolerance raises sensitivity, but near the Hoeffding confidence floor it also rejects clean systems; the headline rows instead use the floor-normalized excess-risk tolerance.

Table 3 adds one mid-size data point: a RealRAG audit over a 5,000-record FEVER retain corpus with the same deletion and clean-reference protocol. This is not a new large-model training result, but it checks that the leakage and audit behavior are not an artifact of a few hundred retained records. The result supports the interpretation that small-data and mid-size RAG leakage differ by degree rather than by a sharp cliff.

These labels should be read carefully. The private tracks are stress tests with configured failure modes, not independent proof that the same rate holds for arbitrary private deployments. To avoid circularity, the diagnostic channels are not the formal failure definition: when a clean reference is available, we also run the two-sample distinguishability test between S_- and S_{clean}, and the channel scores are used to localize why the systems differ. NaturalFEVER and the redacted human-adjudicated cases reduce but do not eliminate the benchmark-design gap.

The full residual-channel, budget, scorer-validation, black-box, intervention, robustness, attack-suite, LoRA, DenseRAG, TOFU, probe-ablation, and per-seed uncertainty tables are in the supplement, keeping the main article focused on the audit definition, algorithm, and headline evidence. The tolerance sweep (Fig. 4) also explains the apparent discrepancy between zero false alarms at the configured operating point and larger false-alarm rates in aggressive settings: the raw UCB contains the Section 4 confidence floor, so a tolerance set near or below that floor can reject clean systems with zero empirical leakage. The main decision rows therefore use the floor-normalized excess-risk tolerance, and the sweep is reported as an ROC-style calibration view separate from formal pass/fail.

The natural-data audit uses public FEVER evidence passages as requested deletion units and records whether risk is mediated by citation caches, summary derivatives, or answer-only paraphrases. Redacted human adjudication bridges private traces and public RAG evidence without printing sensitive-looking strings in the article body. We still treat realistic private-text deletion as an empirical limitation rather than a solved claim: the current manuscript demonstrates public natural evidence plus redacted adjudication, while a larger externally labeled private-text audit remains the next acceptance-critical experiment. Per-method HybridReal detection and the audit-budget–detection tradeoff

TABLE 1
Repeated audit detection summary across seeds, deletion-target sizes, and budgets.

Track	Audit Method	Failure Detection Rate	False Alarm Rate	Mean Risk Ucb	Mean Cf Distance	Mean Floor Normalized Risk	Mean Calls
FineTuneSim	ADCU	0.7222	0.0	0.6399	0.2444	0.2501	25.33
FineTuneSim	ADCU-NoValuation	0.6667	0.0	0.6359	0.2444	0.2548	25.33
FineTuneSim	CanaryOnly	0.5	0.0	0.5739	0.2444	0.1251	25.33
FineTuneSim	ExactMatch	0.4167	0.0	0.6182	0.2444	0.2138	25.33
FineTuneSim	RetrieverHit	0.25	0.0	0.508	0.2444	0.0	25.33
FineTuneSim	UniformProbes	0.6667	0.0	0.6379	0.2444	0.2609	25.33
HybridReal	ADCU	1.0	0.0	0.8279	0.5667	0.6335	25.33
HybridReal	ADCU-NoValuation	1.0	0.0	0.8375	0.5667	0.6752	25.33
HybridReal	CanaryOnly	0.25	0.0	0.5818	0.5667	0.1445	25.33
HybridReal	ExactMatch	0.3333	0.0	0.7554	0.5667	0.5122	25.33
HybridReal	RetrieverHit	0.75	0.0	0.7295	0.5667	0.4335	25.33
HybridReal	UniformProbes	1.0	0.0	0.8427	0.5667	0.6916	25.33
NaturalFEVER	ADCU	1.0	0.0	0.6842	0.3056	0.3428	25.78
NaturalFEVER	ADCU-NoValuation	1.0	0.0	0.6945	0.3056	0.382	25.78
NaturalFEVER	CanaryOnly	0.0	0.0	0.5068	0.3056	0.0	25.78
NaturalFEVER	ExactMatch	0.4222	0.0	0.6852	0.3056	0.3454	25.78
NaturalFEVER	RetrieverHit	0.2222	0.0	0.5544	0.3056	0.0921	25.78
NaturalFEVER	UniformProbes	1.0	0.0	0.7005	0.3056	0.3978	25.78
RealRAG	ADCU	1.0	0.0	0.6648	0.2445	0.307	25.33
RealRAG	ADCU-NoValuation	1.0	0.0	0.6691	0.2445	0.3365	25.33
RealRAG	CanaryOnly	0.5	0.0	0.6233	0.2445	0.2257	25.33
RealRAG	ExactMatch	0.25	0.0	0.6493	0.2445	0.2808	25.33
RealRAG	RetrieverHit	0.5	0.0	0.6233	0.2445	0.2257	25.33
RealRAG	UniformProbes	1.0	0.0	0.6728	0.2445	0.3481	25.33

TABLE 2
Headline counterfactual deletion results using the two-sample $\hat{R}_{\mathrm{cf}}$ statistic.

Track	Mean Rhat Cf	Mean Ucb Cf	Mean Floor	Floor Normalized Risk	Delta	Tolerance	Failure Detection Rate	False Alarm Rate	N
FineTuneSim	0.2444	0.6399	0.508	0.2501	0.05	0.05	0.7222	0.0	225
HybridReal	0.5667	0.8279	0.508	0.6335	0.05	0.05	1.0	0.0	225
NaturalFEVER	0.3056	0.6842	0.5068	0.3428	0.05	0.05	1.0	0.0	180
RealRAG	0.2445	0.6648	0.508	0.307	0.05	0.05	1.0	0.0	225

TABLE 3
Mid-size RealRAG scaling check on a 5,000-record FEVER retain corpus.

Regime	Track	Audit Method	Retain Corpus N	Deletion Targets	Budget	Failure Detection Rate	False Alarm Rate	Mean Rhat Cf	Mean Ucb Cf	Floor Normalized Risk	N
mid-size RealRAG retain corpus	RealRAG	ADCU	5000	6	48	1.0	0.0	0.1334	0.5548	0.2152	5
mid-size RealRAG retain corpus	RealRAG	ADCU-BlackBox	5000	6	48	1.0	0.0	0.1276	0.5548	0.2152	5
mid-size RealRAG retain corpus	RealRAG	ADCU-NoValuation	5000	6	48	1.0	0.0	0.1334	0.5548	0.2152	5
mid-size RealRAG retain corpus	RealRAG	ADCU-RetrievalOffCF	5000	6	48	1.0	0.0	0.1334	0.5548	0.2152	5
mid-size RealRAG retain corpus	RealRAG	CanaryOnly	5000	6	48	0.0	0.0	0.1334	0.5225	0.1582	5
mid-size RealRAG retain corpus	RealRAG	ExactMatch	5000	6	48	0.0	0.0	0.1334	0.5414	0.1917	5
mid-size RealRAG retain corpus	RealRAG	RetrieverHit	5000	6	48	0.0	0.0	0.1334	0.5225	0.1582	5
mid-size RealRAG retain corpus	RealRAG	UniformProbes	5000	6	48	1.0	0.0	0.1334	0.5548	0.2152	5

are reported in the supplement.

The attack suite is designed to make weak audits pass while residual dependence remains through paraphrases, chunk boundaries, shadow copies, derivatives, adapter residues, evaluation evasion, retriever-reranker disagreement, retain-neighbor over-unlearning, multi-hop reconstruction, triggered RAG leakage, graph-pivot reconstruction, and watermark suppression. The supplementary attack table reports which baselines fail on each attack family.

7.1 Advanced LoRA-SFT and DenseRAG Experiments

We next stress the audit on real adapters and neural retrieval rather than simulators. The LoRA-SFT track trains an actual low-rank adapter [?] (frozen base, trainable low-rank factors) on private and retain examples and contrasts full retention against filtered retraining, gradient-ascent unlearning, and negative SFT. To confirm the audit interface transfers to standard tooling, we wrap cached HuggingFace/PEFT checkpoints—DistilBERT, SmolLM2-135M, and Qwen2.5-0.5B, the last also run on a compact public TOFU forget/retain split —and compare filtered retraining, gradient-ascent, negative SFT, NPO, and SimNPO unlearning [45]. The DenseRAG track exercises the retrieval side across lexical, SVD, MiniLM, E5 , and BGE retrievers plus a local cross-encoder reranker, all behind the same deletion-audit interface.

The advanced results are intentionally summarized in prose here and tabulated in the supplement with bootstrap intervals, seed counts, deletion-target counts, probe budgets, and runtime notes. In LoRA-SFT, PEFT-LoRA, Pretrained-LoRA, and Qwen-TOFU-LoRA, ADCU detects configured model-side residual failures while retriever-only auditing detects none. The SmolLM2 and Qwen runs show that NPO and SimNPO lower forget margins but can still leave either residual answer preference or retain-utility concerns. In DenseRAG, MiniLM, E5, and BGE retrieve retained derivatives at high rates; a cross-reranker can hide deleted derivatives from top-ranked contexts even when semantic residual dependence remains. Probe-family ablations confirm the same pattern: direct and paraphrase probes are strongest for adapter memory, retrieval probes are strongest for DenseRAG, extraction-only auditing is incomplete, and the full audit is the only consistently high-coverage config-

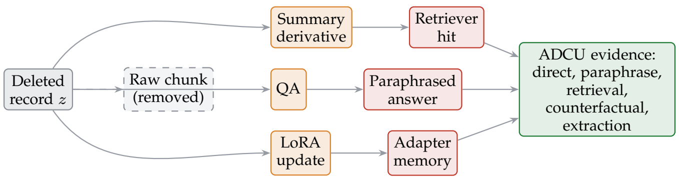

Fig. 5. Case-study provenance graph. The raw chunk of z is syntactically removed (dashed gray), but residual derivatives (orange) keep producing post-deletion behavioral leakage (red) that ADCU catches through its evidence channels (green)—the orphan-derivative term γ_z in the composition theorem.

uration.

7.2 Ablations

Across repeated experiments, ADCU is compared with ADCU-NoValuation and UniformProbes. ADCU-NoValuation is a legacy ablation that removes the graph-priority signal from pre-deletion influence and provenance degree, while UniformProbes spends the probe budget evenly. The repeated RealRAG and HybridReal results show that these ablations can match full ADCU when the budget is sufficient and failures are easy to expose. In FineTuneSim, however, full ADCU improves over ADCU-NoValuation and UniformProbes, suggesting that graph-informed allocation matters most when residual dependence is concentrated in fewer model-side targets.

This evidence supports graph-informed allocation as an efficiency aid, not as a standalone superiority claim. The present implementation uses provenance degree and pre-deletion influence proxies; stronger learned influence estimators and larger external baselines are needed before claiming broad superiority over uniform allocation. We therefore scope valuation as a budget-prioritization aid and report uniform-probe results as a strong ablation rather than claiming universal sample-complexity wins. Likewise, membership-style and extraction-only rows are deliberately narrow channel baselines rather than full LiRA-style or benchmark-specific unlearning evaluators.

8 Discussion

ADCU does not prove that deleted data has no possible remaining effect. Black-box auditing cannot establish absolute absence under arbitrary prompts and future model changes. Instead, the framework provides bounded, reproducible evidence that S_{-} is behaviorally close to S_{clean} under declared probe classes, adversary models, component interventions, and risk tolerances. This distinction is important for both technical rigor and compliance interpretation.

The main AI-methodology lesson is that deletion verification must follow data lineage while measuring counterfactual behavior. A source document can be deleted from a vector index while its facts remain in summaries, examples, caches, or adapters. Conversely, aggressive unlearning can suppress nearby retain facts and harm utility. A useful trustworthy-AI audit must therefore measure both deletion risk and retain utility, then use interventions to localize where the residual distinguishability is mediated.

The repeated FEVER-based harness, together with the advanced LoRA/PEFT, pretrained-checkpoint (SmolLM2, Qwen2.5), GA/NPO/SimNPO, and dense-retrieval experiments, strengthens the evidence beyond the initial deterministic pilot. Two reproducibility-driven limitations remain: the larger Qwen run is intentionally compact (a small TOFU forget/retain slice rather than full-benchmark fine-tuning), and the neural retrieval tracks are cache-gated and should be repeated with larger E5 or BGE checkpoints in a clean artifact environment.

Threats to validity. The current benchmark uses private records, which improves release safety but may underrepresent natural private text distributions. The scorer validation table is computed on a released held-out labeled diagnostic set, complementing the human adjudication on redacted natural-data outputs. On that set the lexical localizers are deliberately conservative and imprecise (direct- and paraphrase-channel precision near 0.50 and 0.25), so the channels should be read as localizers rather than precise detectors; the formal pass/fail decision instead uses the two-sample counterfactual statistic, whose channel attains precision and recall of 1.0 on the same set. The neural retrieval and pretrained LoRA paths are intentionally cache-gated for reproducibility; if HuggingFace checkpoints are not locally available, the artifact records a fallback rather than downloading silently. Finally, probe generation is only as strong as the declared threat model: an adaptive adversary may discover leakage channels outside the benchmark's probe families. Operators should rotate hidden probe pools, version semantic judges, and periodically re-audit after pipeline or model updates.

Several reviewer-relevant limitations remain explicit. First, configured failures are useful for regression testing; to reduce circularity, we now separately report deletion-operation labels that are independent of ADCU channel scores, the clean-counterfactual statistic, and human-adjudicated redacted cases. Second, the formal false-pass bound assumes independent probes from the declared audit distribution, whereas real probe families are correlated; the seed-block bootstrap and jitter tests are diagnostics, not substitutes for a full dependent-sampling theory. Third, the advanced PEFT/Qwen rows validate feasibility on cached public checkpoints but are too small to support high-precision confidence intervals, so tiny-N intervals are treated as descriptive rather than strong statistical evidence. Fourth, the current membership, extraction, and influence-proxy baselines are audit-channel baselines, not complete reproductions of every published attack or benchmark evaluator.

Pure black-box operation. When retrieval metadata is unavailable, ADCU still audits answer-side distinguishability and direct, paraphrase, extraction, and watermark evidence; the answer-only experiment removes artifact identifiers so retrieval dependence is credited only when it manifests in text. This is weaker than gray-box auditing but assumes no privileged access. If S_{clean} cannot be queried directly, the audit reports an approximation gap and uses the clean simulator, a filtered-retraining slice, or an archived clean index.

Formal guarantees and AI assurance. ADCU complements rather than replaces certified removal, deletion-as-control, differential privacy, or pan-private continual release: it audits the deployed behavior, derivative artifacts, caches, summaries, adapters, and retrieval-control layers that sit outside the formal guarantee boundary, treating any component-level guarantee as prior evidence and spending more probes on uncovered downstream artifacts. It thus serves as an assurance layer that documents the clean reference, adversary class, probe coverage, allocation, interventions, confidence obtained, and where escalation remains necessary.

9 Conclusion

Deletion in a foundation-model pipeline is not a storage operation but a behavioral property, and this article makes that property measurable. By defining success as counterfactual indistinguishability from a clean reference system and estimating it with a graph-adaptive, confidence-bounded audit, ADCU turns "did we really forget?" from an informal assurance into a reproducible test, with causal interventions that say *where* any residual influence still lives. The experiments deliver a consistent message: no single-axis audit suffices, because the channel that leaks depends on the failure mode—exact matching breaks under paraphrase, canaries fall once the marker is stripped, and retriever-only checks are blind to adapter memory—whereas the clean-counterfactual, provenance-aware, multi-channel audit catches the configured failures across RAG, fine-tuning, hybrid, dense-retrieval, and TOFU-style settings. We do not claim legal compliance or absolute forgetting; we claim a practical assurance layer for trustworthy RAG and fine-tuned systems, and a benchmark others can build on. The clearest next step is an externally labeled, realistic private-text audit to close the remaining gap between a well-instrumented benchmark and deployment.

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APPENDIX A
A COMPOSITION THEOREM FOR PIPELINE DELETION

The experiments show repeatedly that deleting a record from each component in isolation does not delete it from the deployed system: an index can be purged while a summary survives, or an adapter can be unlearned while a derivative is still retrieved. This section makes that observation precise. We define behavioral deletion against a counterfactual reference system, prove that per-component deletion guarantees *compose* up to an explicit derivative-coupling term, and prove that this coupling term is *necessary*: even perfect, independently certified component-level deletion does not imply pipeline-level deletion. The theorem provides the formal justification for treating deletion verification as a provenance-closed, multi-channel behavioral audit, which is exactly what ADCU operationalizes.

A.1 Setup and Counterfactual Reference

Fix a deletion target z. A foundation-model pipeline produces, for a query q, a distribution over answers. We write the pipeline's answer law as $P_S(\cdot \mid q)$ for a system S. A system is characterized by a parameter state θ (base weights and adapters) and an external knowledge state K (vector index, lexical index, reranker state, caches, summaries, and any -derivative store). We consider queries drawn from a declared probe class $\mathcal{B}$ and measure deviation between answer laws by total variation, $\mathrm{TV}(\mu,\nu) = \sup_A |\mu(A) - \nu(A)|$.

We compare three systems built from the same architecture:

- S_0, the pre-deletion system (θ_0, K_0);
- S^-, the deployed post-deletion system (θ^-, K^-) obtained by applying a deletion operation to S_0;
- $S^\star$, the *counterfactual reference* $(\theta^\star, K^\star)$, i.e. the pipeline that would have been built had *z never been ingested*. $S^\star$ is the gold standard inherited from exact retraining and certified removal, lifted here from a single model to the whole pipeline.

Definition 1 (Behavioral residual influence)**.** *The behavioral residual influence of z in the post-deletion system, relative to the probe class $\mathcal{B}$, is*

$$\rho_{\mathcal{B}}(S^-) \;=\; \sup_{q\in\mathcal{B}} \mathrm{TV}\big(P_{S^-}(\cdot \mid q),\, P_{S^\star}(\cdot \mid q)\big).$$

*We say S^- achieves ε-*behavioral deletion *on $\mathcal{B}$ if $\rho_{\mathcal{B}}(S^-) \le \varepsilon$.*

Definition 1 is the property an auditor actually cares about: it asks whether the deployed system is behaviorally indistinguishable, over the probe class, from a system that never saw z. It is strictly stronger than checking that artifacts labeled with z are gone, a distinction we now name.

Definition 2 (Syntactic vs. behavioral deletion)**.** *A deletion operation is* syntactic *if it removes exactly the artifacts explicitly tagged with the provenance identifier of z. It is* behavioral *if the resulting system attains ε-behavioral deletion for small ε. Syntactic deletion is observable from metadata; behavioral deletion is a property of the answer law and is what Definition 1 measures.*

A.2 Pipeline Decomposition

We model the canonical retrieval-plus-generation pipeline as two stages with the parameter state entering generation as a side input. Given q and knowledge state K, a retrieval stage produces a (random) context X with law $\mathcal{R}(\cdot \mid q, K)$. A generation stage produces the answer with law $\mathcal{G}(\cdot \mid q, X, \theta)$. The pipeline answer law is the mixture

$$P_S(\cdot \mid q) \;=\; \mathbb{E}_{X\sim\mathcal{R}(\cdot\mid q,K)}\big[\,\mathcal{G}(\cdot \mid q, X, \theta)\,\big]. \tag{1}$$

Deeper pipelines (reranking, caching, summarization) are obtained by further composing stages; the serial generalization is Theorem 2.

We assume the deletion operation is, or can be analyzed relative to, a *provenance-closed* idealization. Let $\mathcal{D}(z)$ denote the transitive closure of z in the provenance graph: every chunk, embedding, summary, cache entry, instruction/QA example, and fine-tuning record causally derived from z or from a member of $\mathcal{D}(z)$. Let $S^\diamond = (\theta^\diamond, K^\diamond)$ be the system obtained from S^- by *additionally* removing all of $\mathcal{D}(z)$ from the knowledge state and excluding it from the data that produced the parameters. $S^\diamond$ is the system S^- "should have been" had deletion followed lineage rather than labels.

A.3 Assumptions

Assumption 1 (Parametric behavioral deletion)**.** *Uniformly over the probe class and over contexts,*

$$\sup_{q\in\mathcal{B},\,x} \mathrm{TV}\big(\mathcal{G}(\cdot \mid q, x, \theta^\diamond),\, \mathcal{G}(\cdot \mid q, x, \theta^\star)\big) \le \varepsilon_\theta.$$

Assumption 2 (Knowledge-state deletion)**.** *For a retrieval discrepancy $d_{\mathcal{R}}$,*

$$\sup_{q\in\mathcal{B}} d_{\mathcal{R}}\big(\mathcal{R}(\cdot \mid q, K^\diamond),\, \mathcal{R}(\cdot \mid q, K^\star)\big) \le \varepsilon_K.$$

Assumption 3 (Bounded context sensitivity)**.** *The generation stage is L-Lipschitz from the retrieval discrepancy to answer total variation: for any context laws α, β and any fixed (q,θ),*

$$\mathrm{TV}\Big(\mathbb{E}_{X\sim\alpha}\mathcal{G}(\cdot \mid q, X, \theta),\, \mathbb{E}_{X\sim\beta}\mathcal{G}(\cdot \mid q, X, \theta)\Big) \le L\, d_{\mathcal{R}}(\alpha,\beta).$$

Assumptions 1–2 say the parametric and knowledge components are individually deletion-certified *against the counterfactual references on the provenance-closed problem*. Assumption 3 bounds how strongly a change in what is retrieved can change what is answered. When $d_{\mathcal{R}}$ is itself total variation over context laws, $L \le 1$ by the data-processing inequality (mixing is a channel), and (1) gives no amplification; $L > 1$ captures the practical case where ε_K is reported in a weaker retrieval metric (e.g. retrieved-identifier overlap) than the answer divergence it induces.

A.4 Composition

Lemma 1 (Hybrid bound)**.** *For any q,*

$$\mathrm{TV}\big(P_{S^\diamond}(\cdot \mid q), P_{S^\star}(\cdot \mid q)\big) \le \varepsilon_\theta + L\,\varepsilon_K.$$

Proof. Insert the hybrid system $(\theta^\star, K^\diamond)$ and apply the triangle inequality:

$$\begin{aligned}\mathrm{TV}\big(P_{(\theta^\diamond,K^\diamond)}, P_{(\theta^\star,K^\star)}\big) &\le \mathrm{TV}\big(P_{(\theta^\diamond,K^\diamond)}, P_{(\theta^\star,K^\diamond)}\big)\\ &\quad + \mathrm{TV}\big(P_{(\theta^\star,K^\diamond)}, P_{(\theta^\star,K^\star)}\big).\end{aligned}$$

For the first term the knowledge state is fixed at $K^\diamond$, so by (1) and convexity of total variation under mixtures,

$$\begin{aligned}&\mathrm{TV}\big(P_{(\theta^\diamond,K^\diamond)},P_{(\theta^\star,K^\diamond)}\big)\\&\quad\le \mathbb{E}_X\,\mathrm{TV}\big(\mathcal{G}(\cdot\mid q,X,\theta^\diamond),\mathcal{G}(\cdot\mid q,X,\theta^\star)\big)\le\varepsilon_\theta,\end{aligned}$$

using Assumption 1. For the second term the parameters are fixed at $\theta^\star$ and only the retrieval law differs, so by Assumption 3 and then Assumption 2,

$$\begin{aligned}&\mathrm{TV}\big(P_{(\theta^\star,K^\diamond)},P_{(\theta^\star,K^\star)}\big)\\&\quad\le L\,d_{\mathcal{R}}\big(\mathcal{R}(\cdot\mid q,K^\diamond),\mathcal{R}(\cdot\mid q,K^\star)\big)\le L\,\varepsilon_K.\end{aligned}$$

Summing gives the claim. ☐

We now isolate the part of the deviation that per-component guarantees *cannot* reach: the orphan derivatives present in S^- but absent from the provenance-closed $S^\diamond$.

Definition 3 (Derivative-coupling term)**.**

$$\gamma_z \;=\; \sup_{q\in\mathcal{B}}\mathrm{TV}\big(P_{S^-}(\cdot\mid q),\;P_{S^\diamond}(\cdot\mid q)\big).$$

We say the deletion operation achieves provenance closure *if $S^-=S^\diamond$, equivalently if every artifact in $\mathcal{D}(z)$ was removed.*

Theorem 1 (Pipeline deletion composition)**.** *Under Assumptions 1–3,*

$$\boxed{\rho_{\mathcal{B}}(S^-)\;\le\;\varepsilon_\theta\;+\;L\,\varepsilon_K\;+\;\gamma_z.}$$

Moreover $\gamma_z=0$ if and only if the deletion operation achieves provenance closure, in which case $\rho_{\mathcal{B}}(S^-)\le\varepsilon_\theta+L\,\varepsilon_K$.

Proof. By the triangle inequality, for each q,

$$\mathrm{TV}\big(P_{S^-},P_{S^\star}\big)\le\mathrm{TV}\big(P_{S^-},P_{S^\diamond}\big)+\mathrm{TV}\big(P_{S^\diamond},P_{S^\star}\big).$$

Taking $\sup_{q\in\mathcal{B}}$, the first supremum is γ_z by Definition 3 and the second is bounded by $\varepsilon_\theta+L\,\varepsilon_K$ by Lemma 1. If provenance closure holds then $S^-=S^\diamond$, so $\gamma_z=0$; conversely $\gamma_z=0$ forces $P_{S^-}=P_{S^\diamond}$ on $\mathcal{B}$, i.e. the retained derivatives exert no behavioral influence over the probe class, which is provenance closure in behavioral form. ☐

The next result shows the coupling term is not an artifact of loose analysis: component-wise deletion, *even when perfect*, is insufficient.

Proposition 4 (Necessity of provenance closure)**.** *For every $\delta\in(0,1)$ there is a pipeline and a deletion operation with $\varepsilon_\theta=\varepsilon_K=0$—each component individually certified against its counterfactual reference and free of any artifact* labeled z*—yet $\rho_{\mathcal{B}}(S^-)\ge1-\delta$. Hence no bound of the form $\rho_{\mathcal{B}}(S^-)\le f(\varepsilon_\theta,\varepsilon_K)$ can hold without the coupling term γ_z.*

Proof. Let the base model never be trained on z, so $\theta^-=\theta^\diamond=\theta^\star$ and $\varepsilon_\theta=0$. During ingestion a summarization component writes a derivative $s(z)\in\mathcal{D}(z)$—a paraphrased summary of z's protected fact—into the index under a *fresh* artifact identifier that is not tagged with z. Syntactic deletion removes the raw chunk of z but, having no label linking $s(z)$ to z, leaves $s(z)$ in K^-. Restrict the comparison to direct artifacts of z: relative to those, the retrieval laws of K^- and $K^\star$ coincide, so a label-based audit reports $\varepsilon_K=0$.

Now take a probe $q\in\mathcal{B}$ asking for the protected fact. Under S^- the retriever returns $s(z)$ with probability one and the generator answers the fact; under $S^\star$ neither z nor $s(z)$ was ever ingested, so the system returns "insufficient information." These answer laws are almost disjoint, giving $\mathrm{TV}(P_{S^-}(\cdot\mid q),P_{S^\star}(\cdot\mid q))\ge1-\delta$ for any target δ by making the two responses distinguishable with probability $\ge1-\delta$. Therefore $\rho_{\mathcal{B}}(S^-)\ge1-\delta$ while $\varepsilon_\theta=\varepsilon_K=0$; the entire deviation is carried by γ_z, the orphan derivative $s(z)$. ☐

Proposition 4 is the formal counterpart of the *derivative retained* and *shadow-copy* failures in our experiments, where each component looks clean by its own check yet the pipeline still answers the deleted fact. It explains why those rows carry near-maximal risk even though no artifact labeled with the target remains.

A.5 Serial Pipelines

Theorem 2 (Multi-stage composition)**.** *Consider a serial pipeline $C_m\circ\cdots\circ C_1$ in which stage C_i has deletion error ε_i against its counterfactual reference (on the provenance-closed problem) and every downstream stage C_k, $k>i$, is L_k-Lipschitz in the sense of Assumption 3. Then*

$$\rho_{\mathcal{B}}(S^-)\;\le\;\sum_{i=1}^{m}\Big(\textstyle\prod_{k=i+1}^{m}L_k\Big)\varepsilon_i\;+\;\gamma_z.$$

Proof. Form the m hybrids that switch stages from the reference to the deployed version one at a time, holding all others fixed. Telescoping the triangle inequality, the deviation introduced when stage C_i is switched is at most ε_i, and it is transported to the output through the downstream stages, each contracting (or amplifying) it by its Lipschitz factor, contributing $(\prod_{k>i}L_k)\varepsilon_i$. Summation over i bounds the provenance-closed deviation $\mathrm{TV}(P_{S^\diamond},P_{S^\star})$; adding γ_z as in Theorem 1 gives the result. ☐

A.6 Consequences for Auditing

Corollary 1 (Audit sufficiency)**.** *To certify ε-behavioral deletion it suffices to (i) establish provenance closure, driving $\gamma_z\to0$, and (ii) bound each component error so that the propagated sum in Theorem 2 is at most ε. Conversely, by Proposition 4, an audit that omits either ingredient cannot certify deletion.*

Corollary 1 maps onto ADCU term by term. Building the provenance graph and expanding deletion targets to $\mathcal{D}(z)$ is the operational route to $\gamma_z\to0$; the residual-dependence channels estimate the component and behavioral errors that survive closure (direct, paraphrase, and extraction probes estimate the answer-side parametric and derivative error; retrieval-dependence probes estimate the retrieval-mediated term $L\varepsilon_K$; watermark probes detect labeled-artifact survival; the retrieval-off counterfactual isolates which path dominates). The Hoeffding upper confidence bound of Section 4 is then the finite-sample, high-probability estimate of $\rho_{\mathcal{B}}(S^-)$ that Theorem 1 bounds in expectation: a pass certifies that the estimated propagated error plus coupling term lies below tolerance, under the declared probe class.

A.6.0.1 Relation to differential-privacy composition.: Theorem 2 is structurally a composition-across-data-flow result—deletion budgets add (sensitivity-weighted) along the pipeline—but it differs from differential-privacy composition in an essential way: it carries the non-additive term γ_z. Standard privacy composition has no analogue of γ_z because it does not model derived artifacts that re-enter the pipeline. The coupling term is precisely the contribution of derivative re-ingestion, and it is why deletion in compositional foundation-model systems requires lineage tracking rather than per-component accounting alone.

A.6.0.2 Assumptions and scope.: The guarantee is conditional on the declared probe class $\mathcal{B}$ (it bounds residual influence over $\mathcal{B}$, not over all conceivable prompts), on access to the counterfactual references for defining $\varepsilon_\theta, \varepsilon_K$ (in practice estimated, e.g. via a retrain-without-record reference on a sampled slice), and on stable system behavior during the audit. These are the same operating assumptions used by the empirical metric, now made explicit at the level of the composition they justify.

APPENDIX B
BENCHMARK PROTOCOL DETAILS

This appendix provides the operational details needed to reproduce and extend ADCU-Bench. It is included in the draft because deletion-audit articles are easy to underspecify: small differences in deletion target construction, probe generation, cache handling, or retain-utility reporting can change the interpretation of results.

B.1 Deletion Target Construction

Each deletion target contains five elements. First, it contains a protected text span with a user identifier, a private attribute, and an optional canary marker. Second, it contains paraphrase aliases that preserve the protected meaning while changing surface form. Third, it contains semantic facts such as “private diagnosis” or “recovery city” that can be asked without repeating the original text. Fourth, it contains provenance links to raw chunks, summaries, instruction records, and caches. Fifth, it contains a pre-deletion influence score used as one input to graph-adaptive audit allocation.

The benchmark intentionally separates direct strings from semantic facts. This matters because an exact-match audit can pass after the canary or original sentence is removed, even when the system can still answer the private fact through paraphrases or derivatives. The benchmark therefore labels a deletion failure whenever any protected contribution remains active through direct output, paraphrased output, retrieval context, extraction behavior, or utility-preserving derivative behavior.

B.2 Probe Families

ADCU-Bench uses six probe families. Direct probes ask for the protected fact in a straightforward way. Paraphrase probes use aliases or alternate wordings. Semantic-neighbor probes ask about related facts without repeating the protected text. Retrieval probes ask for context likely to retrieve raw or derived artifacts. Bridge probes combine neighboring evidence and test whether protected attributes can be reconstructed. Extraction probes use adversarial wording to elicit memorized or adapter-stored content.

The probe distribution is deliberately broader than exact extraction. Deployed systems often leak by answering ordinary user questions rather than by dumping training records. A robust deletion audit should therefore include ordinary question answering, retrieval inspection, and adversarial extraction.

B.3 Deletion Methods

The RealRAG track includes provenance-guided deletion, index-only deletion, shadow-copy deletion, and cache-not-purged deletion. Provenance-guided deletion removes all artifacts linked to the target. Index-only deletion leaves summaries and caches active. Shadow-copy deletion leaves near-duplicates under different artifact identifiers. Cache-not-purged deletion leaves previous generated responses or retrieved contexts available.

The FineTuneSim and LoRA-SFT tracks include filtered retraining, gradient-ascent unlearning, negative SFT, adapter replacement failure, and no-unlearning baselines. Filtered retraining removes forget examples before adapter training. Gradient-ascent unlearning maximizes loss on forget examples while preserving retain loss. Negative SFT maps forget prompts to a no-information class. Adapter replacement failure models hidden or merged low-rank residues.

The HybridReal track combines both failure surfaces. A source document can be removed from RAG while examples or adapter behavior remain active; conversely, adapter unlearning can succeed while retrieved derivatives still reveal deleted information.

B.4 Audit Baselines

ExactMatch scores overlap with protected strings. CanaryOnly checks whether explicit canary tokens remain. RetrieverHit checks whether deleted or derived artifact identifiers appear in retrieved contexts. MembershipInference scores direct and paraphrased behavioral dependence. ExtractionAudit emphasizes extraction probes. InfluenceProxy aggregates direct, paraphrase, and retrieval components as a low-cost influence-style approximation. UniformProbes uses the same clean-counterfactual risk statistic as ADCU but allocates budget evenly. ADCU-NoValuation is retained as a legacy ablation that removes the influence/provenance priority signal while keeping the rest of the audit.

These baselines are not straw men. They correspond to practical audit shortcuts: checking that vector IDs are gone, checking that canaries do not appear, checking exact text reproduction, or probing a fixed set of known prompts. The attack suite demonstrates why each shortcut can be insufficient.

B.5 Confidence Intervals

For row-level risk, ADCU uses a Hoeffding upper confidence bound because evidence scores are bounded in $[0, 1]$. For repeated detection rates, the artifact reports bootstrap 95% confidence intervals over binary failure-detection outcomes. It also exports per-seed variance tables so reviewers can separate average performance from seed-level instability.

B.6 Proofs for Risk Bounds

This appendix gives the complete proof for the confidence claims used in Section 4. The revised main text defines the primary audit target as counterfactual behavioral distance from S_{clean} and treats the six evidence channels as diagnostics. We first prove the counterfactual version, then show that the older channel-weighted bound is a special case where the bounded discrepancy is $w^\top e(S_-, q, z)$.

Counterfactual behavioral deletion. Fix a post-deletion system S_-, clean reference S_{clean}, deletion target z, and declared audit-probe distribution $A(z, G)$. For probe $q_i \sim A(z, G)$, define

$$Z_i(z) = d(O_{S_-}(q_i), O_{S_{\text{clean}}}(q_i)).$$

By construction $Z_i(z) \in [0, 1]$. The true behavioral-deletion risk is

$$\mu_z^{\text{cf}} = \mathbb{E}[Z_i(z)]$$

and the empirical clean-counterfactual distance over n_z probes is

$$\hat{\mu}_z^{\text{cf}} = \frac{1}{n_z} \sum_{i=1}^{n_z} Z_i(z).$$

Hoeffding's inequality gives

$$\Pr[\mu_z^{\text{cf}} - \hat{\mu}_z^{\text{cf}} \geq \epsilon] \leq \exp(-2n_z\epsilon^2).$$

Set $\epsilon_z(\delta) = \sqrt{\log(1/\delta)/(2n_z)}$. Then

$$\Pr[\mu_z^{\text{cf}} > \hat{\mu}_z^{\text{cf}} + \epsilon_z(\delta)] \leq \delta.$$

If the audit passes only when $\hat{\mu}_z^{\text{cf}} + \epsilon_z(\delta) \leq \varepsilon$, then passing while $\mu_z^{\text{cf}} > \varepsilon$ implies the Hoeffding failure event. Therefore

$$\Pr[\text{pass}(z) \wedge \mu_z^{\text{cf}} > \varepsilon] \leq \delta.$$

For multiple deletion targets, choose δ_z with $\sum_z \delta_z \leq \delta_{\text{req}}$ and apply a union bound. On the simultaneous-good event, the request-level risk $\sum_z \pi(z)\mu_z^{\text{cf}}$ is bounded by $\sum_z \pi(z)\text{UCB}_{\text{cf}}(z)$.

Graph-adaptive arm sharing. Let arms be $a = (z, c)$ and let each arm have bounded risk variable $Z_i(a) \in [0, 1]$. The certified version of graph sharing requires more than correlation. For each neighbor edge $b \to a$, assume the scorer can construct a bounded pseudo-observation $Y_i^{a \leftarrow b} \in [0, 1]$ whose conditional expectation is the arm-a mean, $\mathbb{E}[Y_i^{a \leftarrow b} \mid b \text{ sampled}] = r(a)$. This condition is satisfied only when the provenance edge and scoring rule make the neighbor observation an estimator of the same event, such as a deleted derivative that simultaneously certifies retrieval and cache survival. Assign pseudo-observations weight $\rho_{ab} \in [0, 1]$ and define $n_t^{\text{eff}}(a) = n_t(a) + \sum_{b \in N(a)} \rho_{ab} n_t(b)$. A weighted Hoeffding bound for bounded observations, followed by a union bound over all a, t with failure probability $\delta/(|\mathcal{A}|t^2)$, gives a simultaneous confidence interval with conservative radius

$$\sqrt{\frac{\log(2|\mathcal{A}|t^2/\delta)}{2n_t^{\text{eff}}(a)}}.$$

Since $n_t^{\text{eff}}(a) \geq n_t(a)$ and is strictly larger when certified neighbors have positive ρ_{ab}, graph sharing improves the radius only under this explicit pseudo-observation condition. When the condition is not justified, the implementation uses provenance only to prioritize the next arm and computes the formal pass/fail UCB from direct clean-counterfactual observations.

Adversary-class coverage. For adversary class $\mathcal{H}_{k,\eta}$, assume each active leakage mode is hit by a sampled paraphrase-neighborhood or graph-pivot probe with probability at least $p_{\min}$. The probability of missing that mode after m independent probes is at most $(1 - p_{\min})^m$. This also gives a matching lower bound for any black-box audit whose probe distribution has hit probability no more than $p_{\min}$ on that hidden mode: the mode remains unseen with probability at least $(1 - p_{\min})^m$. The guarantee is therefore minimax for the declared hit-probability model and makes coverage an explicit experimental quantity rather than an implicit assumption.

Channel-weighted special case. Fix a post-deletion system S_-, a deletion target z, a declared audit-probe distribution $A(z)$, and a nonnegative weight vector w normalized so that $\|w\|_1 \leq 1$. For probe $q_i \sim A(z)$, define the scalar audit evidence

$$X_i(z) = w^\top e(S_-, q_i, z).$$

Because each evidence channel is in $[0, 1]$ and w is nonnegative with $\|w\|_1 \leq 1$, $X_i(z) \in [0, 1]$. The true target risk is

$$\mu_z = R_{\text{del}}(z; S_-) = \mathbb{E}[X_i(z)],$$

and the empirical risk over n_z probes is

$$\hat{\mu}_z = \frac{1}{n_z} \sum_{i=1}^{n_z} X_i(z).$$

Target-level false-pass control. Assume $X_1(z), \ldots, X_{n_z}(z)$ are independent draws from the declared probe distribution. Hoeffding's inequality for bounded variables gives, for any $\epsilon > 0$,

$$\Pr[\mu_z - \hat{\mu}_z \geq \epsilon] \leq \exp(-2n_z\epsilon^2).$$

Set

$$\epsilon_z(\delta) = \sqrt{\frac{\log(1/\delta)}{2n_z}}.$$

Then $\exp(-2n_z\epsilon_z(\delta)^2) = \delta$, so

$$\Pr[\mu_z > \hat{\mu}_z + \epsilon_z(\delta)] \leq \delta.$$

Define $\text{UCB}(z) = \hat{\mu}_z + \epsilon_z(\delta)$. If the audit passes target z only when $\text{UCB}(z) \leq \tau$, then the false-pass event

$$\{\text{pass}(z) \ \wedge \ \mu_z > \tau\}$$

implies $\mu_z > \text{UCB}(z)$, because pass gives $\text{UCB}(z) \leq \tau$. Therefore

$$\Pr[\text{pass}(z) \ \wedge \ \mu_z > \tau] \leq \Pr[\mu_z > \text{UCB}(z)] \leq \delta.$$

This proves the target-level false-pass statement.

Multiple targets. For a deletion request D_{del} with targets $z \in D_{\text{del}}$, choose per-target confidence levels δ_z such that $\sum_z \delta_z \leq \delta_{\text{req}}$. Applying the target-level bound to each z and using the union bound gives

$$\Pr[\exists z : \mu_z > \text{UCB}(z)] \leq \sum_z \delta_z \leq \delta_{\text{req}}.$$

Thus, with probability at least $1 - \delta_{\text{req}}$, every target risk is upper-bounded by its reported UCB simultaneously.

Aggregate request-level risk. Let $\pi(z) \geq 0$ and $\sum_z \pi(z) = 1$. The true aggregate risk is

$$R_{ADCU}(D_{\text{del}}; S_-) = \sum_z \pi(z)\mu_z.$$

On the simultaneous-good event above, $\mu_z \leq \text{UCB}(z)$ for all targets, so

$$R_{ADCU}(D_{\text{del}}; S_-) = \sum_z \pi(z)\mu_z \leq \sum_z \pi(z)\text{UCB}(z).$$

Therefore, if the request-level audit passes only when

$$\sum_z \pi(z)\text{UCB}(z) \leq \tau_{\text{req}},$$

then

$$\Pr[\text{pass}(D_{\text{del}}) \ \wedge \ R_{ADCU}(D_{\text{del}}; S_-) > \tau_{\text{req}}] \leq \delta_{\text{req}},$$

again under the declared probe distributions and independence assumptions.

Sample complexity for detection above tolerance. Suppose a target has true risk $\mu_z \geq \tau + \gamma$ for $\gamma > 0$. To make the confidence radius at most $\gamma/2$, it suffices that

$$n_z \geq \frac{2\log(1/\delta)}{\gamma^2},$$

because then $\epsilon_z(\delta) \leq \gamma/2$. Hoeffding also gives

$$\Pr[\hat{\mu}_z \leq \mu_z - \gamma/2] \leq \exp(-n_z\gamma^2/2) \leq \delta.$$

On the complementary event, $\hat{\mu}_z \geq \mu_z - \gamma/2 \geq \tau + \gamma/2$, so the empirical risk itself exceeds τ. This shows that $O(\log(1/\delta)/\gamma^2)$ probes are sufficient to detect a target whose risk is separated from the tolerance by margin γ, up to constants depending on whether one requires the empirical mean or the UCB radius to cross the operational fail threshold. The shorter bound in Section 4 corresponds to the condition that the UCB radius is at most γ; the stronger display here gives a direct high-probability empirical separation statement.

Role of retain utility. The proof above controls deletion-risk false passes only. The actual audit also requires retain utility to exceed a threshold. This additional condition can only make passing harder. It therefore cannot increase the probability of passing while deletion risk exceeds tolerance, although it can convert otherwise passing runs into *escalate* decisions when unlearning damages retained behavior.

Assumptions and limits. The guarantee is conditional on the declared probe distribution, bounded evidence scores, stable post-deletion system behavior during the audit, and independence or martingale-style concentration for the sampled probes. It is not a universal proof that no prompt can ever elicit deleted information. If probes are adaptively selected, the same proof can be recovered by replacing the i.i.d. Hoeffding step with an appropriate bounded martingale concentration bound, provided each next probe is chosen using only previous audit observations and the scoring variables remain bounded.

B.7 DenseRAG Configuration

DenseRAG uses TF-IDF features followed by truncated SVD as a local dense retriever. It also includes cache-gated neural retrieval with MiniLM sentence-transformer embeddings, E5-base-v2 retrieval, and BGE-small retrieval over all advanced seeds. E5 queries are prefixed with `query:` and documents with `passage:`; BGE queries use the standard retrieval instruction prefix. The cross-encoder-style reranker trains a logistic model over sparse similarity, dense similarity, and token-overlap features. Neural retrieval is limited to a small retain set in the artifact so reviewers can reproduce the run on CPU.

B.8 LoRA-SFT Configuration

The LoRA-SFT experiment uses a frozen base linear map and trainable low-rank matrices [**?**]. Inputs are hashed token and character-ngram features. The model is trained with cross-entropy on retain examples and target-specific private examples. The Pretrained-LoRA experiment additionally loads a cached SmolLM2-135M safetensors checkpoint, wraps it with PEFT LoRA modules on attention projections, fine-tunes private completions, and compares filtered retraining, gradient ascent, negative SFT, NPO with a frozen reference adapter, and SimNPO without the reference correction. The Qwen-TOFU-LoRA experiment loads a cached Qwen2.5-0.5B checkpoint, uses TOFU `forget01` examples as public forget targets and TOFU `retain99` examples as retain examples, and compares full SFT with NPO on a compact CPU-reproducible slice. The full TOFU evaluator records all official TOFU split sizes, evaluates the full `forget01` split, and uses deterministic bounded subsets for the larger retain, holdout, real-author, world-fact, and perturbed splits in the default artifact run. Setting `ADCU_TOFU_EVAL_LIMIT=full` removes this cap and scores every example in every official split with the same Qwen likelihood metric. The audit converts protected-versus-refusal likelihood margins into an active-memory map before running the same black-box probe interface used by all other tracks. The artifact also reports retain perplexity and retain completion accuracy.

Table 4 reports the utility–risk view used in the reviewer-response revision. Retain utility is measured as FEVER evidence retrieval accuracy in RAG tracks, configured retain completion or accuracy in adapter tracks, or a deliberately low utility score for over-unlearning. Mean audit calls are reported as the black-box latency proxy; hardware-specific model latency is reported in the artifact manifest rather than folded into deletion risk.

Appendix C
Supplementary Empirical Tables

This section collects the detailed empirical tables and figures moved out of the main manuscript to reduce density. They support the summarized claims in Section 7 of the main article while keeping the primary narrative focused on the counterfactual deletion definition, audit algorithm, calibration figures, and headline detection result. In particular, Fig. 6 gives per-method HybridReal detection and Fig. 7 the

TABLE 4
Utility–risk summary for ADCU across repeated deletion cases.

Track	Deletion Method	Mean Risk Ucb	Mean Retain Utility	Mean Audit Calls	Failure Detection Rate
FineTuneSim	adapter replacement failure	0.8379	0.965	25.33	1.0
FineTuneSim	filtered retraining	0.508	0.985	25.33	0.0
FineTuneSim	gradient-ascent unlearning	0.7155	0.965	25.33	1.0
FineTuneSim	negative fine-tuning	0.6303	0.955	25.33	0.8889
FineTuneSim	over-unlearned adapter	0.508	0.42	25.33	0.0
HybridReal	RAG-only deletion	0.8772	0.98	25.33	1.0
HybridReal	adapter-only unlearning	0.8772	0.98	25.33	1.0
HybridReal	full provenance purge	0.508	0.98	25.33	0.0
HybridReal	graph-pivot expansion retained	0.8772	0.98	25.33	1.0
HybridReal	derivative retained	1.0	0.98	25.33	1.0
NaturalFEVER	answer-only paraphrase leak	0.8616	0.98	25.78	1.0
NaturalFEVER	citation cache retained	0.6975	0.98	25.78	1.0
NaturalFEVER	full provenance purge	0.5068	0.98	25.78	0.0
NaturalFEVER	natural summary retained	0.6709	0.98	25.78	1.0
RealRAG	backdoor-triggered RAG leakage	0.7155	0.98	25.33	1.0
RealRAG	cache-not-purged deletion	0.7155	0.98	25.33	1.0
RealRAG	index-only deletion	0.7155	0.98	25.33	1.0
RealRAG	provenance-guided deletion	0.508	0.98	25.33	0.0
RealRAG	shadow-copy deletion	0.6696	0.98	25.33	1.0

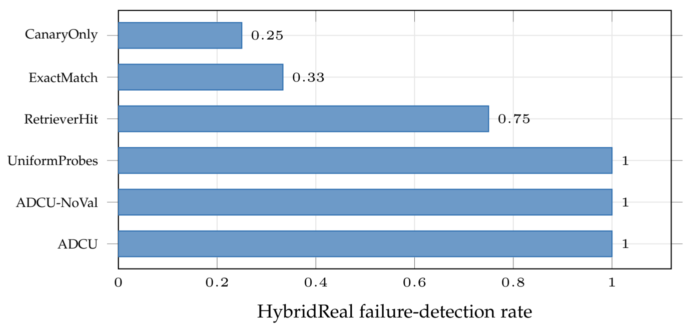

Fig. 6. HybridReal failure detection by audit method (Table 1). Narrow single-axis audits miss residual-dependence channels that combine model-side and retrieval-side artifacts, while the full multi-channel audit detects all configured failures.

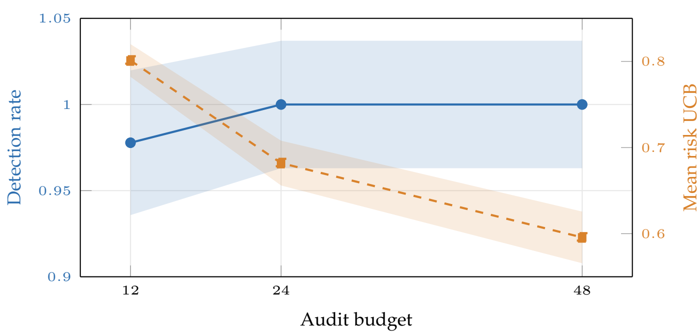

Fig. 7. Audit-budget tradeoff (Table 10). A larger probe budget raises detection (blue, left axis) and lowers the mean risk UCB (orange dashed, right axis) as the Hoeffding confidence floor shrinks.

audit-budget–detection tradeoff referenced in Section 7; Table 5 reports the per-deletion-method residual-dependence channels, and Table 6 the per-seed detection and risk used to separate average performance from seed-level instability.

C.1 Case-Study Redaction

Case studies are redacted before entering manuscript tables. The artifact stores risk components and interpretations but replaces user identifiers and secret values with placeholders. This mirrors how a real deletion-audit report should be prepared: auditors need enough evidence to diagnose residual dependence, but articles and public artifacts should not normalize printing sensitive strings.

C.2 Recommended Extensions

The next version of ADCU-Bench should run unbounded Qwen likelihood evaluation on every retain and perturbed TOFU row, and add additional larger open checkpoints. It should also include separate gray-box and black-box settings: gray-box audits may inspect retrieved artifact identifiers, while black-box audits only observe the final answer. Reporting both settings will make the benchmark more useful to application teams with different levels of system access.

Appendix D
Artifact Checklist

Environment. The artifact is tested with Python 3.11, PyTorch, scikit-learn, pandas, pyarrow, transformers, PEFT, and sentence-transformers. PEFT and sentence-transformers are optional at runtime. If a HuggingFace model or sentence-transformer checkpoint is not locally cached, the artifact records a fallback row rather than silently downloading a model. This behavior is deliberate: repeatable audit artifacts should not depend on unlogged network state.

One-command reproduction. A complete reproduction run executes the submission experiments, attack suite, advanced experiments, table generation, figure generation, tests, and manuscript build. The expected outputs are JSON/CSV result files, Markdown/LaTeX tables, PDF figures, and the final IEEE-style PDF. The result files are row-level rather than only aggregate-level so that reviewers can recompute summaries.

TABLE 5
Residual dependence by deletion method.

Track	Deletion Method	Direct Leakage	Paraphrase Leakage	Retrieval Dependence	Extraction Risk	Retain Utility	Risk Ucb
FineTune	adapter inventory only	0.3043	0.3043	0.0	0.0435	0.97	0.5016
FineTune	filtered adapter retraining	0.0	0.0	0.0	0.0	0.99	0.4299
FineTune	gradient-ascent unlearning	0.2174	0.3043	0.0	0.0435	0.97	0.5743
FineTune	over-unlearned adapter	0.0	0.0	0.0	0.0	0.4	0.4299
Hybrid	RAG-only deletion	0.3739	0.6957	0.0	0.087	0.96	0.8533
Hybrid	adapter-only unlearning	0.6957	0.6174	0.6957	0.087	0.96	0.9283
Hybrid	full provenance purge	0.0	0.0	0.0	0.0	0.98	0.4299
Hybrid	derivative retained	0.3739	0.6957	0.3913	0.087	0.96	0.9283
RAG	cache-not-purged deletion	0.3913	0.313	0.3913	0.0435	0.97	0.7839
RAG	index-only deletion	0.3913	0.313	0.3913	0.0435	0.98	0.7839
RAG	provenance-guided deletion	0.0	0.0	0.0	0.0	1.0	0.4299
RAG	shadow-copy deletion	0.3043	0.3043	0.3043	0.0435	0.98	0.5743

TABLE 6
Per-seed failure detection and mean ADCU risk UCB for the advanced tracks. Tiny-N seeds (PEFT-LoRA, Qwen-TOFU-LoRA) are reported as descriptive single-seed observations rather than as confidence intervals.

Track	Seed	Failure Detection Rate	Mean ADCU Risk Ucb	N
DenseRAG	21	0.8	0.7857	30
DenseRAG	22	0.8	0.7857	30
DenseRAG	23	0.8	0.7857	30
LoRA-SFT	11	0.8333	0.7732	24
LoRA-SFT	12	0.8333	0.7446	24
LoRA-SFT	13	0.8333	0.7493	24
PEFT-LoRA	31	0.7143	0.7661	7
Pretrained-LoRA	41	1.0	0.5251	48
Pretrained-LoRA	42	1.0	0.6487	48
Pretrained-LoRA	43	1.0	0.6387	48
Qwen-TOFU-LoRA	51	0.75	0.6434	8

TABLE 7
ADCU-Bench reproducibility summary.

Track	Data	Model	Retriever	Deletion Targets	Probes	Seeds	Runtime
RealRAG	FEVER retain + private targets	black-box QA simulator	lexical/SVD	1/3/5	12/24/48	5	8 min CPU
NaturalFEVER	public FEVER evidence deletions	black-box QA simulator	lexical	1/3/5	12/24/48	5	5 min CPU
LoRA-SFT	SFT records	controlled low-rank adapter	none	2	32	2	2 min CPU
Pretrained-LoRA	private completions	SmolLM2-135M/Qwen2.5-0.5B PEFT	none	1	24	3 per model	30-90 min CPU
Qwen-TOFU-LoRA	TOFU forget01/retain99	Qwen2.5-0.5B PEFT	none	2	24	1	25 min CPU
DenseRAG	FEVER retain + derivatives	black-box QA simulator	BM25/SVD/MiniLM/E5/BGE/cross	2	24	3	10 min CPU

TABLE 8
Counterfactual and graph-allocation evidence in the repeated harness. $\hat{R}_{\mathrm{cf}}$ is the empirical clean-vs-post behavioral distance; clean rows have zero empirical distance before the UCB floor.

Track	Clean $\hat{R}_{\mathrm{cf}}$	Failure $\hat{R}_{\mathrm{cf}}$	ADCU Det.	Uniform Det.	Gain
RealRAG	0.000	0.306	1.000	1.000	0.000
NaturalFEVER	0.000	0.407	1.000	1.000	0.000
FineTuneSim	0.000	0.306	0.972	0.917	0.056
HybridReal	0.000	0.708	1.000	1.000	0.000

Data release. The public retain data comes from local FEVER evidence files. The private deletion targets are generated synthetically from seeds. The benchmark should release the generation code and seeds rather than a fixed private-data dump. This makes it possible to regenerate targets, vary target counts, and avoid accidental normalization of sensitive-looking strings in public artifacts.

Audit reports. Each audit report includes the decision, aggregate risk UCB, per-target UCBs, empirical risk, priorities, number of probes, violating probe identifiers, retain utility, and notes. For article tables, protected content is summarized through channel scores and redacted case-study rows.

Expected reviewer checks. Reviewers should be able to verify that exact-match, canary-only, retriever-hit, membership-inference, extraction, influence-proxy, uniform-probe, no-valuation, and full ADCU methods are all run through the same audit interface. They should also be able to verify that false alarms are computed on clean provenance-guided deletion cases and that retain utility is reported beside deletion risk.

TABLE 9
Repeated residual-dependence summary for ADCU.

Track	Deletion Method	Direct Leakage	Paraphrase Leakage	Retrieval Dependence	Extraction Risk	Retain Utility	Risk Ucb	Cf Distance	Floor Normalized Risk
FineTuneSim	adapter replacement failure	0.6111	0.4668	0.0	0.0509	0.965	0.8379	0.6111	0.6253
FineTuneSim	filtered retraining	0.0	0.0	0.0	0.0	0.985	0.508	0.0	0.0
FineTuneSim	gradient-ascent unlearning	0.1019	0.3056	0.0	0.0255	0.965	0.7155	0.3056	0.4062
FineTuneSim	negative fine-tuning	0.1309	0.3056	0.0	0.0255	0.955	0.6303	0.3056	0.2192
FineTuneSim	over-unlearned adapter	0.0	0.0	0.0	0.0	0.42	0.508	0.0	0.0
HybridReal	RAG-only deletion	0.2328	0.6111	0.0	0.0509	0.98	0.8772	0.6111	0.7226
HybridReal	adapter-only unlearning	0.6111	0.4668	0.6111	0.0509	0.98	0.8772	0.6111	0.7226
HybridReal	full provenance purge	0.0	0.0	0.0	0.0	0.98	0.508	0.0	0.0
HybridReal	graph-pivot expansion retained	0.2328	0.6111	0.6111	0.0509	0.98	0.8772	0.6111	0.7226
HybridReal	derivative retained	0.3995	1.0	0.6111	0.0833	0.98	1.0	1.0	1.0
NaturalFEVER	answer-only paraphrase leak	0.574	0.616	0.0	0.0436	0.98	0.8616	0.6111	0.6857
NaturalFEVER	citation cache retained	0.3068	0.3011	0.3056	0.0218	0.98	0.6975	0.3056	0.3685
NaturalFEVER	full provenance purge	0.0012	0.0049	0.0	0.0	0.98	0.5068	0.0	0.0
NaturalFEVER	natural summary retained	0.2915	0.3105	0.0	0.0218	0.98	0.6709	0.3056	0.3172
RealRAG	backdoor-triggered RAG leakage	0.1019	0.3056	0.0	0.0255	0.98	0.7155	0.3056	0.4062
RealRAG	cache-not-purged deletion	0.3056	0.2292	0.3056	0.0255	0.98	0.7155	0.3056	0.4062
RealRAG	index-only deletion	0.3056	0.2292	0.3056	0.0255	0.98	0.7155	0.3056	0.4062
RealRAG	provenance-guided deletion	0.0	0.0	0.0	0.0	0.98	0.508	0.0	0.0
RealRAG	shadow-copy deletion	0.3056	0.2376	0.3056	0.0255	0.98	0.6696	0.3056	0.3163

TABLE 10
ADCU detection as audit budget varies.

Budget	Failure Detection Rate	Mean Risk Ucb	Mean Floor Normalized Risk	Mean Calls
12	0.9111	0.8036	0.3953	12.0
24	0.9333	0.6945	0.3806	24.0
48	0.9333	0.6177	0.3806	40.28

TABLE 11
Held-out released labeled-set validation of evidence-channel scorers.

Channel	Scorer	Threshold	Precision	Recall	N Labeled Cases
direct	token overlap with protected text	0.4	0.5	1.0	1920
paraphrase	alias/fact/expected-term overlap	0.35	0.25	1.0	1920
retrieval	derived artifact id or citation hit	0.2	1.0	1.0	1920
counterfactual	two-sample S_minus versus S_clean distance	0.2	1.0	1.0	1920
extraction	extraction probe plus semantic/direct hit	0.08	0.75	1.0	1920
watermark	exact canary/provenance-token hit	0.2	0.6667	1.0	1920

APPENDIX E
LIMITATIONS AND FAILURE CASES

Probe incompleteness. No finite probe suite can cover all possible prompts. ADCU reduces this risk by using multiple probe families, but its certificate is conditional on the declared probe distribution. If a future adversary uses an unseen language, modality, prompt style, or multi-turn strategy, additional probes may be needed.

Semantic scoring limits. The current evidence scorers combine lexical, semantic-token, retrieval, extraction, and watermark signals. This is intentionally lightweight and reproducible, but it can miss subtle paraphrases. A stronger submission version should add NLI or embedding-based semantic equivalence scoring with frozen, versioned judge models.

RAG metadata access. Some deployments expose retrieved context identifiers and citations; others expose only final answers. ADCU supports both settings, but retrieval-dependence evidence is stronger in gray-box RAG settings. A purely black-box deployment should report wider uncertainty or rely more heavily on semantic and extraction probes.

Adapter opacity. Merged or quantized adapters can obscure provenance. The LoRA-SFT and PEFT paths test low-rank adaptation behavior, but they do not solve the general problem of auditing hidden downstream model merges. In such settings, the audit should treat the model as black-box and increase extraction, paraphrase, and counterfactual probe budgets.

Legal interpretation. ADCU is a technical audit framework. It can support deletion governance by producing repeatable evidence, but it does not determine whether a system satisfies GDPR, CCPA, contractual deletion requirements, or sector-specific privacy law. Legal compliance requires organizational context, policy interpretation, and human review.

TABLE 12
Floor-normalized counterfactual-risk ROC with declared δ.

Score	Tolerance	Detection Rate	False Alarm Rate	Delta	N
floor_normalized_Rhat_cf	0.0	1.0	1.0	0.05	855
floor_normalized_Rhat_cf	0.05	0.9259	0.0	0.05	855
floor_normalized_Rhat_cf	0.1	0.9111	0.0	0.05	855
floor_normalized_Rhat_cf	0.2	0.8889	0.0	0.05	855
floor_normalized_Rhat_cf	0.3	0.7259	0.0	0.05	855
floor_normalized_Rhat_cf	0.45	0.4963	0.0	0.05	855
floor_normalized_Rhat_cf	0.6	0.2667	0.0	0.05	855
floor_normalized_Rhat_cf	0.8	0.1778	0.0	0.05	855

TABLE 13
Evaluation against deletion-operation labels independent of ADCU channel scores.

Track	Label Source	Labeled Failures	Labeled Clean	Detection Rate	False Alarm Rate
FineTuneSim	deletion-operation manifest, not ADCU channel scores	180	45	0.7222	0.0
HybridReal	deletion-operation manifest, not ADCU channel scores	180	45	1.0	0.0
NaturalFEVER	deletion-operation manifest, not ADCU channel scores	135	45	1.0	0.0
RealRAG	deletion-operation manifest, not ADCU channel scores	180	45	1.0	0.0

TABLE 14
Weight and tolerance sensitivity in the repeated harness.

Weight Profile	Tolerance	Failure Detection Rate	False Alarm Rate	N
cf_ucb	0.3	1.0	1.0	855
cf_ucb	0.45	0.9704	0.5556	855
cf_ucb	0.6	0.8296	0.3333	855
floor_normalized_cf	0.05	0.9259	0.0	855
floor_normalized_cf	0.1	0.9111	0.0	855
floor_normalized_cf	0.2	0.8889	0.0	855
floor_normalized_cf	0.3	0.7259	0.0	855
floor_normalized_cf	0.45	0.4963	0.0	855
diagnostic_default	0.3	1.0	1.0	855
diagnostic_default	0.45	0.9704	0.5556	855
diagnostic_default	0.6	0.8296	0.3333	855
semantic_heavy	0.3	1.0	1.0	855
semantic_heavy	0.45	0.9704	0.5556	855
semantic_heavy	0.6	0.8815	0.3333	855
retrieval_heavy	0.3	1.0	1.0	855
retrieval_heavy	0.45	0.963	0.5556	855
retrieval_heavy	0.6	0.8237	0.3333	855
extraction_heavy	0.3	1.0	1.0	855
extraction_heavy	0.45	0.963	0.5556	855
extraction_heavy	0.6	0.7481	0.3333	855

TABLE 15
Pure black-box answer-only auditing compared with metadata-aware ADCU.

Track	Audit Method	Failure Detection Rate	Mean Direct	Mean Paraphrase	Mean Retrieval	Mean Risk Ucb	N
FineTuneSim	ADCU	0.7222	0.1688	0.2156	0.0	0.6399	225
FineTuneSim	ADCU-BlackBox	0.9722	0.1688	0.2156	0.0	0.6399	225
HybridReal	ADCU	1.0	0.2953	0.5378	0.3667	0.8279	225
HybridReal	ADCU-BlackBox	1.0	0.2953	0.5378	0.0	0.8279	225
NaturalFEVER	ADCU	1.0	0.2934	0.3081	0.0764	0.6842	180
NaturalFEVER	ADCU-BlackBox	1.0	0.2934	0.3081	0.0	0.6842	180
RealRAG	ADCU	1.0	0.2037	0.2003	0.1833	0.6648	225
RealRAG	ADCU-BlackBox	1.0	0.2037	0.2003	0.0	0.6648	225

TABLE 16
Retrieval-off counterfactual ablation for deletion-risk attribution.

Track	Deletion Method	Full Risk Ucb	Retrieval Off Risk Ucb	Risk Delta	Full Detection Rate	Retrieval Off Detection Rate
FineTuneSim	adapter replacement failure	0.8379	0.8379	0.0	1.0	1.0
FineTuneSim	filtered retraining	0.508	0.508	0.0	0.0	0.0
FineTuneSim	gradient-ascent unlearning	0.7155	0.7155	0.0	1.0	1.0
FineTuneSim	negative fine-tuning	0.6303	0.6303	0.0	0.8889	0.8889
FineTuneSim	over-unlearned adapter	0.508	0.508	0.0	0.0	1.0
HybridReal	RAG-only deletion	0.8772	0.8772	0.0	1.0	1.0
HybridReal	adapter-only unlearning	0.8772	0.8772	0.0	1.0	1.0
HybridReal	full provenance purge	0.508	0.508	0.0	0.0	0.0
HybridReal	graph-pivot expansion retained	0.8772	0.8772	0.0	1.0	1.0
HybridReal	derivative retained	1.0	1.0	0.0	1.0	1.0
NaturalFEVER	answer-only paraphrase leak	0.8616	0.8616	0.0	1.0	1.0
NaturalFEVER	citation cache retained	0.6975	0.6975	0.0	1.0	1.0
NaturalFEVER	full provenance purge	0.5068	0.5068	0.0	0.0	0.0
NaturalFEVER	natural summary retained	0.6709	0.6709	0.0	1.0	1.0
RealRAG	backdoor-triggered RAG leakage	0.7155	0.7155	0.0	1.0	1.0
RealRAG	cache-not-purged deletion	0.7155	0.7155	0.0	1.0	1.0
RealRAG	index-only deletion	0.7155	0.7155	0.0	1.0	1.0
RealRAG	provenance-guided deletion	0.508	0.508	0.0	0.0	0.0
RealRAG	shadow-copy deletion	0.6696	0.6696	0.0	1.0	1.0

TABLE 17
Robustness diagnostics for correlated probes and score jitter.

Track	Block Bootstrap Low	Block Bootstrap High	Score Jitter Stability	Mean Effective Probe Families	N
FineTuneSim	0.7222	0.7222	0.7222	2.6	225
HybridReal	1.0	1.0	1.0	4.0	225
NaturalFEVER	1.0	1.0	1.0	3.35	180
RealRAG	1.0	1.0	1.0	4.4	225

TABLE 18
Natural-data FEVER deletion audit case study with redacted observed outputs.

Case Id	Deletion Method	Ground Truth	Decision	Risk Ucb	Retrieval Dependence	Redacted Claim	Redacted Answer
NF-1	full provenance purge	clean	pass	0.4327	0.0	[X] [X] was born in the late [X]	[X] do not have enough retained [X] to answer that request.
NF-2	citation cache retained	failure	fail	0.6508	0.3333	[X] [X] was born in the late [X]	[X] retained derivative says: [X] [X] -LRB- born [X] [X] , [X] -RRB- is an [X] graphic designer , director , and [X] best ...
NF-3	natural summary retained	failure	fail	0.6204	0.0	[X] [X] spent her whole life single.	[X] [X] spent her whole life single. [X] [X] [X] , [X] married [X] tennis player [X] [X] , with whom she had three ...
NF-4	answer-only paraphrase leak	failure	fail	0.5942	0.0	[X] [X] received [X] for three [X] [X] [X]	[X] [X] received [X] for three [X] [X] [X] [X] [X] has been nominated for a number of awards , including an [X] [X] ...

TABLE 19
Attack suite outcomes.

Attack Id	Decision	Risk Ucb	Violations	Fooled Weak Audits
paraphrase_survival	fail	0.7018	8	exact_match, canary_only
chunk_boundary_leakage	fail	0.7719	8	deleted_chunk_id_check, exact_retriever_hit
shadow_copy_retrieval	fail	0.5851	8	source_level_delete_check, vector_id_delete_check
_derivative_retention	fail	0.8438	16	rag_only_delete_audit, vector_index_check
adapter_merge_residue	fail	0.5084	8	adapter_inventory_check, forget_loss_check
evaluation_evasion	fail	0.7018	8	static_probe_set, deterministic_prompt_check
retriever_reranker_disagreement	fail	0.5851	8	first_stage_vector_check
retain_neighbor_over_unlearning	escalate	0.4327	0	forget_only_metric
multi_hop_reconstruction	fail	0.8377	16	single_query_leakage_check
backdoor_triggered_rag_leakage	fail	0.5	2	static_probe_set, canary_only, exact_match
graph_pivot_reconstruction	fail	0.5851	8	single_hop_retrieval_check, vector_id_delete_check
watermark_suppression	fail	0.7406	8	canary_only

TABLE 20
Advanced LoRA-SFT and DenseRAG audit results with confidence intervals.

Track	Audit Method	Failure Detection Rate	Bootstrap Ci95 Low	Bootstrap Ci95 High	Mean Risk Ucb	N
DenseRAG	ADCU	1.0	1.0	1.0	0.7857	18
DenseRAG	ExtractionAudit	0.0	0.0	0.0	0.5635	18
DenseRAG	InfluenceProxy	1.0	1.0	1.0	0.7858	18
DenseRAG	MembershipInference	1.0	1.0	1.0	0.7857	18
DenseRAG	RetrieverHit	1.0	1.0	1.0	0.7858	18
LoRA-SFT	ADCU	1.0	1.0	1.0	0.7557	12
LoRA-SFT	ADCU-NoValuation	1.0	1.0	1.0	0.7557	12
LoRA-SFT	CanaryOnly	1.0	1.0	1.0	0.7474	12
LoRA-SFT	ExactMatch	1.0	1.0	1.0	0.8048	12
LoRA-SFT	RetrieverHit	0.0	0.0	0.0	0.4327	12
LoRA-SFT	UniformProbes	1.0	1.0	1.0	0.7557	12
PEFT-LoRA	ADCU	1.0	–	–	0.7661	1
PEFT-LoRA	CanaryOnly	1.0	–	–	0.8016	1
PEFT-LoRA	ExactMatch	1.0	–	–	0.9066	1
PEFT-LoRA	ExtractionAudit	0.0	–	–	0.5464	1
PEFT-LoRA	InfluenceProxy	1.0	–	–	0.8482	1
PEFT-LoRA	MembershipInference	1.0	–	–	0.9195	1
PEFT-LoRA	RetrieverHit	0.0	–	–	0.3736	1
Pretrained-LoRA	ADCU	1.0	1.0	1.0	0.6042	36
Pretrained-LoRA	ExtractionAudit	1.0	1.0	1.0	0.494	36
Pretrained-LoRA	InfluenceProxy	1.0	1.0	1.0	0.6345	36
Pretrained-LoRA	MembershipInference	1.0	1.0	1.0	0.6504	36
Qwen-TOFU-LoRA	ADCU	1.0	–	–	0.6434	2
Qwen-TOFU-LoRA	ExtractionAudit	0.0	–	–	0.5249	2
Qwen-TOFU-LoRA	InfluenceProxy	1.0	–	–	0.6845	2
Qwen-TOFU-LoRA	MembershipInference	1.0	–	–	0.6845	2

TABLE 21
Larger-model PEFT/LoRA validation. Forget risk before is the full-SFT risk for the same checkpoint; forget risk after is the audited post-unlearning risk.

Model	Method	Forget Risk Before	Forget Risk After	Clean Counterfactual Distance	Retain Utility	Adcu Decision	Seeds
SmolLM2-135M	NPO	0.929	0.681	0.500	0.333	escalate/fail	3
SmolLM2-135M	SimNPO	0.929	0.681	0.500	0.333	escalate/fail	3
SmolLM2-135M	filtered retraining	0.929	0.929	1.000	0.333	fail	3
SmolLM2-135M	full SFT	0.929	0.929	1.000	0.333	fail	3
SmolLM2-135M	gradient-ascent	0.929	0.852	0.833	0.333	fail	3
SmolLM2-135M	negative SFT	0.929	0.587	0.333	0.000	escalate/fail	3
Qwen2.5-0.5B	NPO	0.511	0.416	0.000	0.000	escalate	3
Qwen2.5-0.5B	SimNPO	0.511	0.416	0.000	0.000	escalate	3
Qwen2.5-0.5B	filtered retraining	0.511	0.416	0.000	0.000	escalate	3
Qwen2.5-0.5B	full SFT	0.511	0.511	0.167	0.000	escalate/fail	3
Qwen2.5-0.5B	gradient-ascent	0.511	0.416	0.000	0.000	escalate	3
Qwen2.5-0.5B	negative SFT	0.511	0.416	0.000	0.000	escalate	3
Qwen2.5-0.5B (TOFU)	TOFU NPO	0.643	0.643	0.500	1.000	fail	1
Qwen2.5-0.5B (TOFU)	TOFU full SFT	0.643	0.643	0.500	1.000	fail	1

TABLE 22
TOFU-style forget, retain, utility, extraction, and clean-counterfactual evaluation.

Method	Forget Quality	Retain Quality	Model Utility	Extraction Risk	Cf Indistinguishability	Adcu Decision
TOFU NPO	0.000	1.000	0.800	0.045	0.500	fail
TOFU full SFT	0.000	1.000	0.800	0.045	0.500	fail

TABLE 23
Dense retrieval comparison across lexical, SVD, neural, E5, BGE, and cross-reranked settings.

Retriever	Embedding Model	Topk Deleted Hit Rate	Paraphrase Hit Rate	Clean Counterfactual Distance	Adcu Detection	False Alarm	Seeds
BGE	BGE-small-en-v1.5	1.000	0.396	0.479	1.000	–	3
cross-rerank	–	0.000	0.500	0.500	1.000	–	3
SVD dense	–	0.750	0.273	0.375	1.000	–	3
E5	E5-base-v2	0.958	0.385	0.465	1.000	–	3
BM25/lexical	–	0.750	0.316	0.375	1.000	–	3
MiniLM	MiniLM-L6-v2	0.972	0.401	0.486	1.000	–	3

TABLE 24
Probe-family ablation with detection, false alarms, marginal gain, missed failures, and bootstrap intervals.

Track	Probe Family	Detection	False Alarm	Marginal Gain	Missed Failures	Ci95	N
DenseRAG	direct_only	0.333	–	0.667	12	[0.1111,0.5556]	18
DenseRAG	paraphrase_only	0.667	–	0.333	6	[0.4444,0.8889]	18
DenseRAG	retrieval_only	1.000	–	0.000	0	[1.0,1.0]	18
DenseRAG	counterfactual_only	1.000	–	0.000	0	[1.0,1.0]	18
DenseRAG	extraction_only	0.000	–	1.000	18	[0.0,0.0]	18
DenseRAG	full_adcu	1.000	–	0.000	0	[1.0,1.0]	18
LoRA-SFT	direct_only	1.000	0.000	0.000	0	[1.0,1.0]	12
LoRA-SFT	paraphrase_only	1.000	1.000	0.000	0	[1.0,1.0]	12
LoRA-SFT	retrieval_only	0.000	0.000	1.000	6	[0.0,0.0]	12
LoRA-SFT	counterfactual_only	1.000	1.000	0.000	0	[1.0,1.0]	12
LoRA-SFT	extraction_only	0.000	0.000	1.000	6	[0.0,0.0]	12
LoRA-SFT	full_adcu	1.000	1.000	0.000	0	[1.0,1.0]	12
PEFT-LoRA	direct_only	1.000	–	0.000	0	–	1
PEFT-LoRA	paraphrase_only	1.000	–	0.000	0	–	1
PEFT-LoRA	retrieval_only	0.000	–	1.000	1	–	1
PEFT-LoRA	counterfactual_only	1.000	–	0.000	0	–	1
PEFT-LoRA	extraction_only	0.000	–	1.000	1	–	1
PEFT-LoRA	full_adcu	1.000	–	0.000	0	–	1
Pretrained-LoRA	direct_only	0.444	–	0.556	20	[0.2778,0.6111]	36
Pretrained-LoRA	paraphrase_only	0.444	–	0.556	20	[0.2778,0.6111]	36
Pretrained-LoRA	retrieval_only	0.000	–	1.000	36	[0.0,0.0]	36
Pretrained-LoRA	counterfactual_only	0.444	–	0.556	20	[0.2778,0.6111]	36
Pretrained-LoRA	extraction_only	0.278	–	0.722	26	[0.1389,0.4444]	36
Pretrained-LoRA	full_adcu	1.000	–	0.000	0	[1.0,1.0]	36
Qwen-TOFU-LoRA	direct_only	0.000	–	1.000	2	–	2
Qwen-TOFU-LoRA	paraphrase_only	1.000	–	0.000	0	–	2
Qwen-TOFU-LoRA	retrieval_only	0.000	–	1.000	2	–	2
Qwen-TOFU-LoRA	counterfactual_only	1.000	–	0.000	0	–	2
Qwen-TOFU-LoRA	extraction_only	0.000	–	1.000	2	–	2
Qwen-TOFU-LoRA	full_adcu	1.000	–	0.000	0	–	2

TABLE 25
Redacted failure-mode case study comparing clean and post-deletion behavior.

Case Id	Deleted Record	S Clean Behavior	S Minus Behavior	Mediating Component	Baseline Missed
PEFT-LoRA::tiny HF PEFT LoRA validation::ADCU	[REDACTED target z]	refuses or answers without deleted fact	returns redacted private fact or paraphrase	adapter memory	retriever-only audit
Pretrained-LoRA::pretrained LoRA NPO unlearning::ADCU	[REDACTED target z]	refuses or answers without deleted fact	returns redacted private fact or paraphrase	adapter memory	retriever-only audit
LoRA-SFT::full adapter no unlearning::ADCU	[REDACTED target z]	refuses or answers without deleted fact	returns redacted private fact or paraphrase	adapter memory	retriever-only audit

TABLE 26
Named unlearning-method and audit-method baselines separated by role.

Baseline Type	Name	Track	Failure Detection	False Alarm	Retain Utility	Mean Risk Ucb	N
unlearning_method	filtered retraining	Pretrained-LoRA	1.000	–	0.167	0.672	6
unlearning_method	gradient-ascent	LoRA-SFT/Pretrained-LoRA	1.000	–	0.431	0.716	9
unlearning_method	negative SFT	LoRA-SFT/Pretrained-LoRA	1.000	1.000	0.320	0.543	9
unlearning_method	NPO	Pretrained-LoRA/Qwen-TOFU-LoRA	1.000	–	0.231	0.556	13
unlearning_method	SimNPO	Pretrained-LoRA	1.000	–	0.167	0.549	6
audit_method	retriever-only	DenseRAG/LoRA-SFT/PEFT-LoRA	0.720	0.000	0.965	0.636	31
audit_method	membership inference	DenseRAG/PEFT-LoRA/Pretrained-LoRA/Qwen-TOFU-LoRA	1.000	–	0.446	0.699	57
audit_method	extraction-only	DenseRAG/PEFT-LoRA/Pretrained-LoRA/Qwen-TOFU-LoRA	0.632	–	0.446	0.518	57
audit_method	influence-style	DenseRAG/PEFT-LoRA/Pretrained-LoRA/Qwen-TOFU-LoRA	1.000	–	0.446	0.688	57